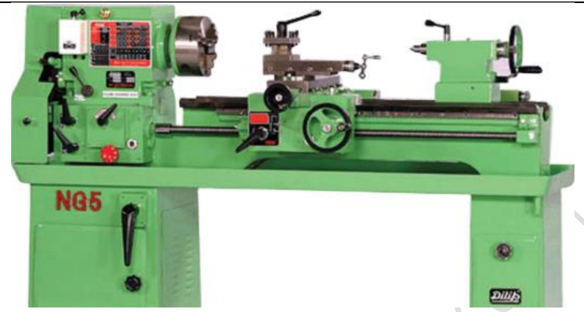
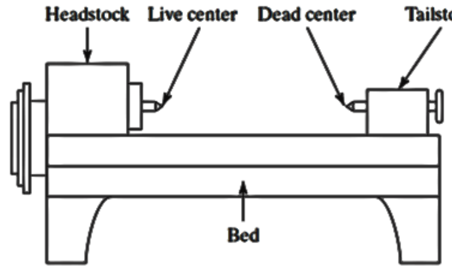
Free vibration of multi-degree freedom systems

Systems that require more than one independent co-ordinates to describe their motion are called multi-degree freedom systems.

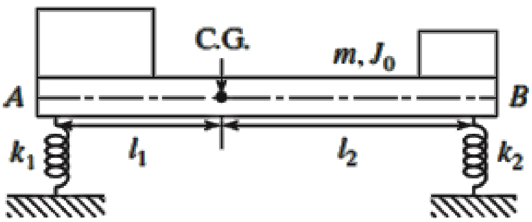
A simplified model of an engine lathe is shown below. For simplified analysis, this can be considered as a rigid body of mass m and mass moment of inertia J_o about its centre of gravity resting on springs of stiffness k_1 and k_2 as shown. The displacement of the system at instant of time can be specified by the vertical displacement $x(t)$ of the C.G. of the mass and the angle $\theta(t)$ denoting the rotation of the mass with respect to its C.G. We can lso use cordinates $x_1(t)$ and $x_2(t)$ representing the vertical displacements of points A and B. Thus, the sytem has two degrees of freedom.

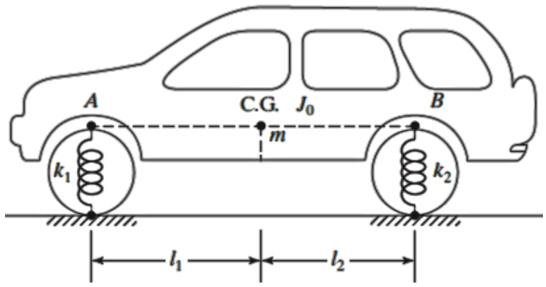




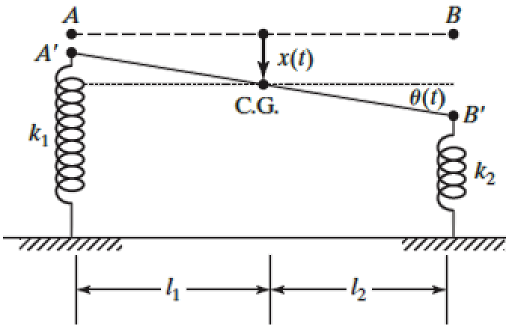
Similarly, a car vibrating in the vertical plane can be modelled as a two degree of freedom sysyem as shown. The car body can be idealised as a bar of mass m and mass moment of inertia J_o about its centre of gravity. Springs of stiffness k_1 and k_2 represent the front and rear suspensions.

There are two equations of motion for a two degree of freedom system, one for each mass (or for each coordinate). The differential equations will be coupled- that is each equation involves all the coordibates. If a harmonic solution is assumed for each coordinate, the equation of motion lead to a frequency equation that will give two natural frequencies for the system. It will vibrate at one of these natural frequencies if it is given a suitable initial excitation.





The amplitudes of the two degrees of freedom or the coordinates are related in a particular manner when it vibrates at one the natural frequencies. That configuration is called normal mode, or principal mode, or natural mode of vibration. If the initial excitation is arbitrary, the resulting free vibration will be a superposition of the two normal modes.



As we have seen above, the configuration of a system can be specified by a set of independent coordinates such as length, angle, or some other physical parameters. Any such set of coordinates is called generalised coordinates.

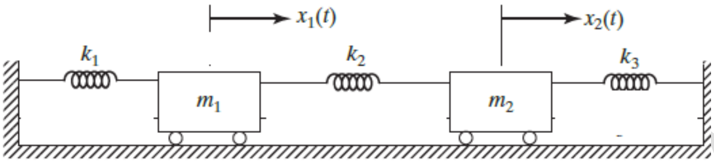
Although the equation of motion of a multi-degree freedom system are generally coupled, we can find a particular set of coordinates such that each equation of motion contains only one coordinate. The equation of motion are then uncoupled and can be solved independently of each other. Such set of coordinates, which leads to an uncoupled set of equations, is called principal coordinates.

Example 1

Find the natural frequency and mode shapes of the two degree of freedom spring mass sytem shown.

Take $m_1 = m_2 = 1\ kg$

$k_1 = k_2 = 2\ kN/m$



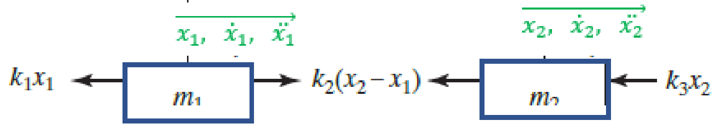
Initial conditions are $x_1(0) = 0.1m;\ x_2(0) = 0.2m$

$\dot{x}_1(0) = 0; \quad \dot{x}_2(0) = 0$

The free body diagrams of the two masses are shown. The equations of motion are

$$m_1\ddot{x}_1 = \Sigma F_{x\ in\ the\ direction\ of\ \ddot{x}_1}$$

$$m_2\ddot{x}_2 = \Sigma F_{x\ in\ the\ direction\ of\ \ddot{x}_2}$$



$$m_1\ddot{x}_1 = k_2(x_2 - x_1) - k_1x_1$$

$$m_1\ddot{x}_1 + (k_1 + k_2)x_1 - k_2x_2 = 0 - - - - (1)$$

$$m_2\ddot{x}_2 = -k_2(x_2 - x_1) - k_3x_2$$

$$m_2\ddot{x}_2 - k_2x_1 + (k_2 + k_3)x_2 = 0 - - - (2)$$

The two equations can be written in the matrix form

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} (k_1 + k_2) & -k_2 \\ -k_2 & (k_2 + k_3) \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} - - - - - - (3)$$

$$[m]\{\ddot{x}\} + [k]\{x\} = \{0\} - - - - (4)$$

This the matrix form of the differential equations.

$[m] = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}$ is known as the mass matrix (inertia matrix) and $[k] = \begin{bmatrix} (k_1 + k_2) & -k_2 \\ -k_2 & (k_2 + k_3) \end{bmatrix}$ is the stiffness matrix. Note that the mass and stiffness matrices will be symmetric. Here, the mass matrix is uncoupled since it is a diagonal matrix and the stiffness matrix is coupled since the off-diagonal elements are not zeroes.

If the mass (inertia) matrix is coupled, it is known as dynamic (inertia) coupling and if the stiffness atrix is coupled, it is called static (elatic) coupling.

Assuming a harmonic solution for displacement of mass m_1

$x_1(t) = X_1 \sin(\omega t + \phi)$ ------- (5)

and for mass m_2

$x_2(t) = X_2 \sin(\omega t + \phi)$--------- (6)

$x_1(t)$ and $x_2(t)$ are the displacements of mass m_1 and m_2 at any time

X_1 and X_2 are the amplitudes of masses, $\quad \omega$ the frequency, and ϕ the phase angle

Combining Eq. (5) and (6)

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \sin(\omega t + \phi) - - - (7)$$

$\{x\}=\{X\}\sin(\omega t+\phi) - - - (8)$

Differentiating the Eq. (5) and (6), we get

$\ddot{x}_1 = -\omega^2 X_1 \sin(\omega t+\phi)$ and $\ddot{x}_2 = -\omega^2 X_2 \sin(\omega t+\phi)$

The first differtial equation (1) now becomes

$$m_1(-\omega^2 X_1 \sin(\omega t+\phi)) + (k_1+k_2)\, X_1 \sin(\omega t+\phi) - k_2\, X_2 \sin(\omega t+\phi) = 0$$

$$\{-\omega^2\, m_1 X_1 + (k_1+k_2)\, X_1) - k_2\, X_2\} \sin(\omega t+\phi) = 0$$

$\{(k_1+k_2)\ -\omega^2\, m_1\}X_1 - k_2\, X_2 = 0 - - - - - - (9)$

$$(k_1+k_2)X_1 - k_2\, X_2 = \omega^2\, m_1 X_1$$

$$\left(\frac{k_1+k_2}{m_1}\right)X_1 - \left(\frac{k_2}{m_1}\right)\, X_2 = \omega^2\, X_1 - - - (10)$$

Similarly, the second differetial equation (2) can be written as

$$m_2(-\omega^2 X_2 \sin(\omega t+\phi)) - k_2\, X_1 \sin(\omega t+\phi) + (k_2+k_3)\, X_2 \sin(\omega t+\phi) = 0$$

$$\{-\omega^2\, m_2 X_2 + (k_2+k_3)\, X_2) - k_2\, X_1\} \sin(\omega t+\phi) = 0$$

$-k_2 X_1 + \{(k_2+k_3) - \omega^2\, m_2\}\, X_2 = 0 - - - - -(11)$

$$-k_2 X_1 + (k_2+k_3)\, X_2 = \omega^2\, m_2 X_2$$

$$-\left(\frac{k_2}{m_2}\right)X_1 + \left(\frac{k_2+k_3}{m_2}\right)\, X_2 = \omega^2\, X_2 - - - -(12)$$

We can write Eq. (10) and (12) as

$$\begin{bmatrix} \left(\frac{k_1+k_2}{m_1}\right) & -\left(\frac{k_2}{m_1}\right) \\ -\left(\frac{k_2}{m_2}\right) & \left(\frac{k_2+k_3}{m_2}\right) \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} \omega^2\, X_1 \\ \omega^2\, X_2 \end{Bmatrix} = \omega^2 \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix}$$

This is now of the form

$$[A]\{X\} = \lambda\{X\} - - - -(13)$$

where $[A] = \begin{bmatrix} \left(\frac{k_1+k_2}{m_1}\right) & -\left(\frac{k_2}{m_1}\right) \\ -\left(\frac{k_2}{m_2}\right) & \left(\frac{k_2+k_3}{m_2}\right) \end{bmatrix}$, $\{X\} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix}$, and $\lambda = \omega^2$

Eq.(5) says that when the vector $\{X\}$ is multiplied by matrix $[A]$ (transformation matrix), the vector $\{X\}$ is stretched or shortened by a factor λ. This is called eigen value problem. λ is known as eigen value and $\{X\}$ is known as the eigen vector.

$$[A]\{X\} - \lambda\{X\} = 0$$

$$([A] - \lambda\{I\})\{X\} = 0$$

In order to have a non-trivial solution for the above equation $|[A] - \lambda\{I\}| = 0$

$$\left| \begin{bmatrix} \left(\frac{k_1+k_2}{m_1}\right) & -\left(\frac{k_2}{m_1}\right) \\ -\left(\frac{k_2}{m_2}\right) & \left(\frac{k_2+k_3}{m_2}\right) \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} \left(\frac{k_1+k_2}{m_1}\right) - \lambda & -\left(\frac{k_2}{m_1}\right) \\ -\left(\frac{k_2}{m_2}\right) & \left(\frac{k_2+k_3}{m_2}\right) - \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 4000-\lambda & -2000 \\ -2000 & 4000-\lambda \end{vmatrix} = 0$$

$$(4000-\lambda)^2 - 4000000 = 0$$

$$\lambda^2 - 8000\lambda + 12000000 = 0$$

$$\lambda = 2000, 6000$$

$$\therefore \boldsymbol{\omega = \pm 44.72\,, \quad \pm 77.46}$$

Another way of looking at the solution

The equations (9) and (11) represent a system of linear homogeneous equations in X_1 and X_2.

$X_1 = 0$ and $X_2 = 0$ are possible solutions for equations (9) and (11), but they are trivial solution as there will not be any vibration. The system of linear homogeneous equations (9) and (11) can be written as matrix equation as given below

$$\begin{bmatrix} (k_1+k_2) - \omega^2\, m_1 & -k_2 \\ -k_2 & (k_2+k_3) - \omega^2\, m_2 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} - - - - - (14)$$

This set of equations in the form $[A]\{X\} = \{0\}$ to have a non trivial solution, the determinant of the coefficient matrix $[A]$ should be zero.

$$\begin{vmatrix} (k_1+k_2) - \omega^2\, m_1 & -k_2 \\ -k_2 & (k_2+k_3) - \omega^2\, m_2 \end{vmatrix} = 0$$

Substituting the values of k and m

$$\begin{vmatrix} 4000-\omega^2 & -2000 \\ -2000 & 4000-\omega^2 \end{vmatrix} = 0$$

$$(4000-\omega^2)^2 - 4000000 = 0$$

$$16000000 - 8000\omega^2 + \omega^4 - 4000000 = 0$$

$\boldsymbol{\omega^4 - 8000\omega^2 + 12000000 = 0 - - - - (15)}$

Substituting $\omega^2 = \lambda$, (6) becomes $\lambda^2 - 8000\lambda + 12000000 = 0$

$$\lambda = 2000, 6000$$

$$\therefore \boldsymbol{\omega = \pm 44.72, \quad \pm 77.46}$$

Taking the positive values for ω,

$$\omega_1 = 44.72\, rad/s, \;\; and\; \omega_2 = 77.46\, rad/s$$

This shows that the system can have non-trivial harmonic solutions of the form given by (5) and (6) when $\boldsymbol{\omega}$ equals to $\omega_1 = 44.72\, rad/s,\;\; and\; \omega_2 = 77.46\, rad/s$. The values of this frequencies are called natural frequencies of the system.

Now, the values of X_1 and X_2 are now to be determined. Their values depend on the values of ω_1 and ω_2. The values of X_1 and X_2 corresponding to ω_1 are denoted as X_1^1 and X_2^1 and those corresponding to ω_2 are denoed as X_1^2 and X_2^2.

Eq.(14) can now be written as

$$\begin{bmatrix}(k_1+k_2)-\omega_1{}^2\, m_1 & -k_2 \\ -k_2 & (k_2+k_3)-\omega_1{}^2\, m_2\end{bmatrix}\begin{Bmatrix}X_1^1\\X_2^1\end{Bmatrix}=\begin{Bmatrix}0\\0\end{Bmatrix}------(16)$$

And

$$\begin{bmatrix}(k_1+k_2)-\omega_2{}^2\, m_1 & -k_2 \\ -k_2 & (k_2+k_3)-\omega_2{}^2\, m_2\end{bmatrix}\begin{Bmatrix}X_1^2\\X_2^2\end{Bmatrix}=\begin{Bmatrix}0\\0\end{Bmatrix}------(17)$$

Equation (16) $\Rightarrow \begin{bmatrix}4000-44.72^2 & -2000 \\ -2000 & 4000-44.72^2\end{bmatrix}\begin{Bmatrix}X_1^1\\X_2^1\end{Bmatrix}=\begin{Bmatrix}0\\0\end{Bmatrix}$

$$\begin{bmatrix}2000 & -2000 \\ -2000 & 2000\end{bmatrix}\begin{Bmatrix}X_1^1\\X_2^1\end{Bmatrix}=\begin{Bmatrix}0\\0\end{Bmatrix}$$

Since this is a set of two homogeneous equations in variables and X_1^1 *and* X_2^1, we can have infinite solutions,or we can express it as the ratio of the variables $r_1=\frac{X_2^1}{X_1^1}$

From the first of the equations we can write; $2000\,X_1^1-2000\,X_2^1=0 \Longrightarrow \frac{X_2^1}{X_1^1}=1=r_1$

From the second of the equations; $-2000\,X_1^1+2000\,X_2^1=0 \Longrightarrow \frac{X_2^1}{X_1^1}=1=r_1$

Equation (17) $\Rightarrow \begin{bmatrix}4000-77.46^2 & -2000 \\ -2000 & 4000-77.46^2\end{bmatrix}\begin{Bmatrix}X_1^2\\X_2^2\end{Bmatrix}=\begin{Bmatrix}0\\0\end{Bmatrix}$

Or $\begin{bmatrix}-2000 & -2000 \\ -2000 & -2000\end{bmatrix}\begin{Bmatrix}X_1^2\\X_2^2\end{Bmatrix}=\begin{Bmatrix}0\\0\end{Bmatrix}$

Again, this is another set of two homogeneous equations in variables and X_1^2 *and* X_2^2, we can have infinite solutions,or we can express it as the ratio of the variables $r_2=\frac{X_2^2}{X_1^2}$

From the first of the equations we can write; $-2000\,X_1^2-2000\,X_2^2=0 \Longrightarrow \frac{X_2^2}{X_1^2}=-1=r_2$

From the second of the equations; $-2000\,X_1^2-2000\,X_2^2=0 \Longrightarrow \frac{X_2^2}{X_1^2}=-1=r_2$

If we choose any value for X_1^1, we will get a corresponding value for $X_2^1=r_1\,X_1^1$

Similarly, for any value of X_1^2, the corresponfing value for $X_2^2=r_2\,X_1^2$

Let us assume, $X_1^1=1$, and $X_1^2=1$

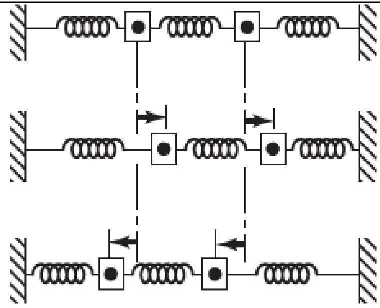
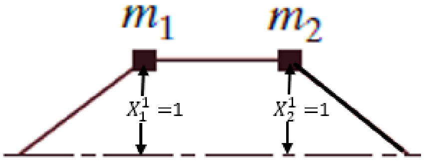
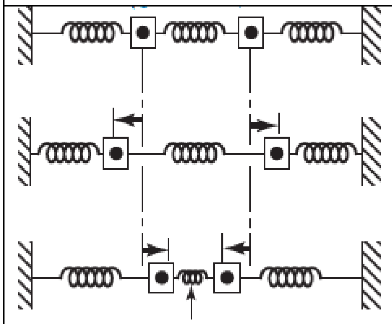
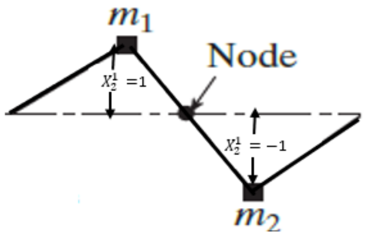
Then, $X_2^1=r_1\,X_1^1=1$ and $X_2^2=r_2\,X_1^2=-1$

The normal modes of vibation corresponding to ω_1 and ω_2 can be expressed as

$$\vec{X}^1=\begin{Bmatrix}X_1^1\\X_2^1\end{Bmatrix}=\begin{Bmatrix}X_1^1\\r_1X_1^1\end{Bmatrix}$$

$$\vec{X}^2=\begin{Bmatrix}X_1^2\\X_2^2\end{Bmatrix}=\begin{Bmatrix}X_1^2\\r_2X_1^2\end{Bmatrix}$$

The vectors $\vec{X}^1$ and $\vec{X}^2$, wich denote the normal modes of vibration are also known as the modal vectors (Eigen vectors) of the system

	 Ist Mode shape Both masses move in phase	Normal modes or modal vectors $\{X\}^1 = \begin{Bmatrix} X_1{}^1 \\ X_2{}^1 \end{Bmatrix} = X_1{}^1 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$
 Node	 IInd Mode shape Both masses move out of phase	Normal modes or modal vectors $\{X\}^2 = \begin{Bmatrix} X_1{}^2 \\ X_2{}^2 \end{Bmatrix} = X_1{}^2 \begin{Bmatrix} 1 \\ -1 \end{Bmatrix}$

For the Ist mode,the free vibration solution Eq. (8) can be written

$\{\vec{x}\}^1 = \{\vec{X}\}^1 \sin(\omega_1 t + \phi_1)$

$\begin{Bmatrix} x_1^1 \\ x_2^1 \end{Bmatrix} = \begin{Bmatrix} X_1^1 \\ X_2^1 \end{Bmatrix} \sin(\omega_1 t + \phi_1) = \begin{Bmatrix} X_1^1 \\ r_1 X_1^1 \end{Bmatrix} \sin(\omega_1 t + \phi_1)$. This is the first mode.

Similarly for the 2nd mode, Eq. (8) becomes

$\{\vec{x}\}^2 = \{X\}^2 \sin(\omega_2 t + \phi_2)$

$\begin{Bmatrix} x_1^2 \\ x_2^2 \end{Bmatrix} = \begin{Bmatrix} X_1^2 \\ X_2^2 \end{Bmatrix} \sin(\omega_2 t + \phi_2) = \begin{Bmatrix} X_1^2 \\ r_2 X_1^2 \end{Bmatrix} \sin(\omega_2 t + \phi_2)$. This is the second mode.

$\vec{X}^1, \vec{X}^2, \phi_1, \phi_2$ can be found out from the initial conditions.

If we give the initial conditions as below, the first normal mode of vibration will be initiated.

Initial displacement $\Rightarrow x_1(t=0) = X_1^1$ and $x_2(t=0) = r_1 X_1^1$

Initial velocity$\Rightarrow \dot{x}_1(t=0) = 0$ and $\dot{x}_2(t=0) = 0$

To get the second normal mode of vibration, the initial cnditions should be as below.

Initial displacement $\Rightarrow x_1(t=0) = X_1^2$ and $x_2(t=0) = r_2 X_1^2$

Initial velocity$\Rightarrow \dot{x}_1(t=0) = 0$ and $\dot{x}_2(t=0) = 0$

For all other initial conditions, both modes will be excited. The resulting motion can be obtained by the linear combination (superposition) of the two normal modes. The general solution can be written as

$$\vec{x}(t) = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$
$$= \vec{x}^1(t) + \vec{x}^2(t)$$

$$= \{\vec{X}^1\}\sin(\omega_1 t+\phi_1) + \{\vec{X}^2\}\sin(\omega_2 t+\phi_2) = \begin{Bmatrix} X_1^1 \\ X_2^1 \end{Bmatrix} sin(\omega_1 t+\phi_1) + \begin{Bmatrix} X_1^2 \\ X_2^2 \end{Bmatrix} sin(\omega_2 t+\phi_2)$$

$$= \begin{Bmatrix} X_1^1 \\ r_1 X_1^1 \end{Bmatrix}\sin(\omega_1 t+\phi_1). + \begin{Bmatrix} X_1^2 \\ r_2 X_1^2 \end{Bmatrix}\sin(\omega_2 t+\phi_2).$$

$$= \begin{Bmatrix} X_1^1\ \sin(\omega_1 t+\phi_1) + X_1^2\ \sin(\omega_2 t+\phi_2) \\ r_1 X_1^1\ \sin(\omega_1 t+\phi_1) + r_2 X_1^2 \sin(\omega_2 t+\phi_2) \end{Bmatrix} --- (18)$$

The initial displacement of the masses; $x_1(t=0) = x_1(0)$ and $x_2(t=0) = x_2(0)$

The initial velocity of the masses; $\dot{x}_1(t=0) = \dot{x}_1(0)$ and $\dot{x}_2(t=0) = \dot{x}_2(0)$

We have the displacement equations

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1^1\ \sin(\omega_1 t+\phi_1) + X_1^2\ \sin(\omega_2 t+\phi_2) \\ r_1 X_1^1\ \sin(\omega_1 t+\phi_1) + r_2 X_1^2 \sin(\omega_2 t+\phi_2) \end{Bmatrix} ---- (19)$$

Differentiating with respect to time

$$\begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{Bmatrix} = \begin{Bmatrix} \omega_1 X_1^1\ \cos(\omega_1 t+\phi_1) + \omega_2 X_1^2\ \cos(\omega_2 t+\phi_2) \\ \omega_1 r_1 X_1^1\ \cos(\omega_1 t+\phi_1) + \omega_2 r_2 X_1^2 \cos(\omega_2 t+\phi_2) \end{Bmatrix} ---- (20)$$

Substituting the initial conditions of initial displacement and initial velocity, we get four equations;

$x_1(0) = X_1^1\ \sin(\phi_1) + X_1^2\ \sin(\phi_2) -- (21)$ ⊗ $\dot{x}_1(0) = \omega_1 X_1^1\ \cos(\phi_1) + \omega_2 X_1^2\ \cos(\phi_2) -- (22)$

$x_2(0) = r_1 X_1^1\ \sin(\phi_1) + r_2 X_1^2 \sin(\phi_2) -- (23)$ ⊗ $\dot{x}_2(0) = \omega_1 r_1 X_1^1\ \cos(\phi_1) + \omega_2 r_2 X_1^2\ \cos(\phi_2) -- (24)$

Eq. (21)⇒

$$X_1^2\ \sin(\phi_2) = x_1(0) - X_1^1\ \sin(\phi_1) --- (25)$$

Substituting in Eq. (23)⇒

$$x_2(0) = r_1 X_1^1\ \sin(\phi_1) + r_2\ (x_1(0) - X_1^1\ \sin(\phi_1))$$

$$\Rightarrow\ x_2(0) - r_2 x_1(0) = (r_1 - r_2)\ X_1^1\ \sin(\phi_1)$$

$$X_1^1\ \sin(\phi_1) = \frac{x_2(0) - r_2 x_1(0)}{(r_1 - r_2)} --- (26)$$

Substituting in Eq. (25)⇒

$$X_1^2\ \sin(\phi_2) = x_1(0) - \frac{x_2(0) - r_2 x_1(0)}{(r_1 - r_2)} = \frac{r_1\ x_1(0) - x_2(0)}{(r_1 - r_2)}$$

$$X_1^2\ \sin(\phi_2) = \frac{r_1\ x_1(0) - x_2(0)}{(r_1 - r_2)} --- (29)$$

Eq. (22)⇒

$$X_1^2\ \cos(\phi_2) = \dot{x}_1(0) - \frac{\omega_1 X_1^1\ \cos(\phi_1)}{\omega_2} --- (27)$$

Substituting in Eq. (24)⇒

$$\dot{x}_2(0) = \omega_1 r_1 X_1^1\ \cos(\phi_1) + \dot{x}_1(0) - \omega_2 r_2 \left(\frac{\omega_1 X_1^1\ \cos(\phi_1)}{\omega_2}\right)$$

$$\Rightarrow\ \omega_1(r_1 - r_2)(X_1^1\ \cos(\phi_1) = \dot{x}_2(0) - \dot{x}_1(0)$$

$$\Rightarrow\ X_1^1\ \cos(\phi_1) = \frac{\dot{x}_2(0) - \dot{x}_1(0)}{\omega_1(r_1 - r_2)} --- (28)$$

Substituting in Eq. (27)⇒

$$X_1^2\ \cos(\phi_2) = \dot{x}_1(0) - \frac{\omega_1}{\omega_2}\left(\frac{\dot{x}_2(0) - \dot{x}_1(0)}{\omega_1(r_1 - r_2)}\right)$$

$$X_1^2\ \cos(\phi_2) = \dot{x}_1(0) - \left(\frac{\dot{x}_2(0) - \dot{x}_1(0)}{\omega_2(r_1 - r_2)}\right) --- (30)$$

$X_1^1 \sin(\phi_1) = \dfrac{x_2(0) - r_2 x_1(0)}{(r_1 - r_2)}$	$X_1^1 \cos(\phi_1) = \dfrac{\dot{x}_2(0) - \dot{x}_1(0)}{\omega_1(r_1 - r_2)}$
Squaring and adding these two equations (26) and (28), we can find the value of X_1^1 $X_1^1 = \sqrt{\left\{\dfrac{x_2(0) - r_2 x_1(0)}{(r_1 - r_2)}\right\}^2 + \left\{\dfrac{\dot{x}_2(0) - \dot{x}_1(0)}{\omega_1(r_1 - r_2)}\right\}^2} \quad --(31)$	Dividing (26) by (28) $\tan\phi_1 = \omega_1 \left(\dfrac{x_2(0) - r_2 x_1(0)}{\dot{x}_2(0) - \dot{x}_1(0)}\right) --(32)$

$X_1^2 \sin(\phi_2) = \dfrac{r_1\, x_1(0) - x_2(0)}{(r_1 - r_2)} --- (29)$	$X_1^2 \cos(\phi_2) = \dot{x}_1(0) - \left(\dfrac{\dot{x}_2(0) - \dot{x}_1(0)}{\omega_2(r_1 - r_2)}\right) --(30)$
Squaring and adding these two equations (29) and (30), we can find the value of X_1^2 $X_1^2 = \sqrt{\left\{\dfrac{r_1\, x_1(0) - x_2(0)}{(r_1 - r_2)}\right\}^2 + \left\{\dot{x}_1(0) - \left(\dfrac{\dot{x}_2(0) - \dot{x}_1(0)}{\omega_2(r_1 - r_2)}\right)\right\}} --(33)$	Dividing (29) by (30) $\tan\phi_2 = \omega_2 \left(\dfrac{r_1\, x_1(0) - x_2(0)}{\omega_2 \dot{x}_1(0)(r_1 - r_2) - (\dot{x}_2(0) - \dot{x}_1(0))}\right) --(34)$

Initial conditions given in the problem are

$$x_1(0) = 0.1m; \qquad x_2(0) = 0.2m$$

$$\dot{x}_1(0) = 0; \qquad \dot{x}_2(0) = 0$$

$$\tan\phi_1 = \omega_1 \left(\frac{x_2(0) - r_2 x_1(0)}{\dot{x}_2(0) - \dot{x}_1(0)}\right) = \infty \Rightarrow \phi_1 = \frac{\pi}{2}$$

$$\tan\phi_2 = \omega_2 \left(\frac{r_1\, x_1(0) - x_2(0)}{\omega_2 \dot{x}_1(0)(r_1 - r_2) - (\dot{x}_2(0) - \dot{x}_1(0))}\right) = \infty \Rightarrow \phi_2 = \frac{\pi}{2}$$

Substituting the values of $\phi_1 = \frac{\pi}{2} = \phi_2$ in (33) and (34)

$$X_1^1 = \frac{x_2(0) - r_2 x_1(0)}{(r_1 - r_2)} = \frac{0.2 - (-1) \times 0.1}{(1 - (-1))} = 0.15$$

$$X_1^2 = \frac{r_1\, x_1(0) - x_2(0)}{(r_1 - r_2)} = \frac{(1) \times (0.1) - 0.2}{(1 - (-1))} = -0.05$$

The general solution is

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1^1 \sin(\omega_1 t + \phi_1) + X_1^2 \sin(\omega_2 t + \phi_2) \\ r_1 X_1^1 \sin(\omega_1 t + \phi_1) + r_2 X_1^2 \sin(\omega_2 t + \phi_2) \end{Bmatrix}$$

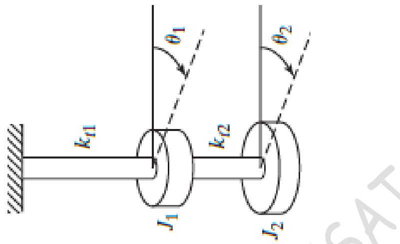
$$= \begin{Bmatrix} 0.15 \sin(44.72t + \frac{\pi}{2}) - 0.05 \sin(77.46t + \frac{\pi}{2}) \\ 0.15 \sin(44.72t + \frac{\pi}{2}) + 0.05 \sin(77.46t + \frac{\pi}{2}) \end{Bmatrix}$$

$$x_1 = 0.15 \cos 44.72t - 0.05 \cos 76.46t$$

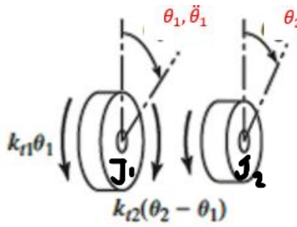
$$x_2 = 0.15 \cos 44.72t + 0.05 \cos 76.46t$$

Example 2

Find the natural frequency and mode shapes of the two degree of freedom torsional system shown for $J_1 = 2\ kg.m^2,\ J_2 = 4\ kg.m^2, k_{t1} = k_{t2} = 10000\ Nm/rad$



The free body diagram of the system



The equation of motion for the first rotor

$$J_1\ddot{\theta}_1 = \Sigma M_{in\ the\ direction\ of\ \ddot{\theta}_1}$$
$$= k_t(\theta_2 - \theta_1) - k_t\theta_1$$
$$J_1\ddot{\theta}_1 + 2k_t\theta_1 - k_t\theta_2 = 0 ----(1)$$

The equation of motion for the second rotor

$$J_2\ddot{\theta}_2 = \Sigma M_{in\ the\ direction\ of\ \ddot{\theta}_2}$$
$$= -k_t(\theta_2 - \theta_1)$$
$$J_2\ddot{\theta}_2 - k_t\theta_1 + k_t\theta_2 = 0 ----(2)$$

Combining the equations (1) and (2) in the matrix form

$$\begin{bmatrix} J_1 & 0 \\ 0 & J_2 \end{bmatrix}\begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} 2k_t & -k_t \\ -k_t & k_t \end{bmatrix}\begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} ----(3)$$

$$[J]\{\ddot{\theta}\} + [k]\{\theta\} = \{0\} ------(4)$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}\begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} 20000 & -10000 \\ -10000 & 10000 \end{bmatrix}\begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Assuming harmonic solution $\{\theta\}=\{\Theta\}\sin(\omega t + \phi) --- (4)$

where $\{\theta\}$ =angular displacements of rotors 1 and 2= $\begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix}$

$\{\Theta\}$ =amplitudes of angular displacements of rotors 1 and 2 = $\begin{Bmatrix} \Theta_1 \\ \Theta_2 \end{Bmatrix}$

Differentiating with respect to time, $\{\ddot{\theta}\} = -\omega^2\{\Theta\}\sin(\omega t + \phi)$

Eq. (4) becomes $-[J]\omega^2\{\Theta\}\sin(\omega t + \phi) + [k]\{\Theta\}\sin(\omega t + \phi) = \{0\}$

$$\{-\omega^2[J] + [k]\}\{\Theta\}\sin(\omega t + \phi) = \{0\} --- (5)$$

$$\{-\omega^2[J] + [k]\}\{\Theta\} = \{0\}$$

$$\left\{-\omega^2\begin{bmatrix} J_1 & 0 \\ 0 & J_2 \end{bmatrix} + \begin{bmatrix} 2k_t & -k_t \\ -k_t & k_t \end{bmatrix}\right\}\begin{Bmatrix} \Theta_1 \\ \Theta_2 \end{Bmatrix} = \{0\}$$

$$\begin{bmatrix} 2k_t - \omega^2 J_1 & -k_t \\ -k_t & k_t - \omega^2 J_2 \end{bmatrix}\begin{Bmatrix} \Theta_1 \\ \Theta_2 \end{Bmatrix} = \{0\} ----(6)$$

The condition for the system of linear homogeneous equations to have a non-trivial solutions, determinanat of the coefficient matrix=0

$$\begin{vmatrix} 2k_t - \omega^2 J_1 & -k_t \\ -k_t & k_t - \omega^2 J_2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 20000 - 2\omega^2 & -10000 \\ -10000 & 10000 - 4\omega^2 \end{vmatrix} = 0$$

$$(20000 - 2\omega^2)(10000 - 4\omega^2) - 10^8 = 0$$

$$2 \times 10^8 - 8 \times 10^4\omega^2 - 2 \times 10^4\omega^2 + 8\omega^4 - 10^8 = 0$$

$$8\omega^4 - 10 \times 10^4\omega^2 + 10^8 = 0$$

Substituting $\omega^2 = \lambda$

$$8\lambda^2 - 10 \times 10^4\lambda + 10^8 = 0$$

$$\lambda = \frac{10 \times 10^4 \pm \sqrt{100 \times 10^8 - 32 \times 10^8}}{16} = \frac{10 \times 10^4 \pm 8.25 \times 10^4}{16} = 1093.75 \ and \ 11406.25$$

$$\therefore \omega = 33.1\frac{rad}{s};\ 106.8\frac{rad}{s}$$

The natural frequecies of the two rotor system are $\omega_1 = 33.1\, rad/s$ and $\omega_2 = 106.8\, rad/s$

When $\omega = \omega_1 = 33.1\, rad/s$ (First Mode of vibration), the equation (6) becomes

$$\begin{bmatrix} 2k_t - {\omega_1}^2 J_1 & -k_t \\ -k_t & k_t - {\omega_1}^2 J_2 \end{bmatrix} \begin{Bmatrix} \Theta_1^1 \\ \Theta_2^1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} 20000 - 2 \times 33.1^2 & -10000 \\ -10000 & 10000 - 4 \times 33.1^2 \end{bmatrix} \begin{Bmatrix} \Theta_1^1 \\ \Theta_2^1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$17808.78\ \Theta_1^1 - 10000\ \Theta_2^1 = 0 \quad \Rightarrow\Rightarrow \quad \frac{\Theta_2^1}{\Theta_1^1} = 1.78$$

and $-10000\ \Theta_1^1 + 5617.56\ \Theta_2^1 = 0 \Rightarrow\Rightarrow \quad \frac{\Theta_2^1}{\Theta_1^1} = 1.78$

$$r_1 = 1.78$$

If $\Theta_1^1 = 1$, then $\Theta_2^1 = 1.78$

When $\omega = \omega_2 = 106.8\, rad/s$ (Second Mode of vibration), the equation (6) becomes

$$\begin{bmatrix} 2k_t - {\omega_2}^2 J_1 & -k_t \\ -k_t & k_t - {\omega_2}^2 J_2 \end{bmatrix} \begin{Bmatrix} \Theta_1^2 \\ \Theta_2^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

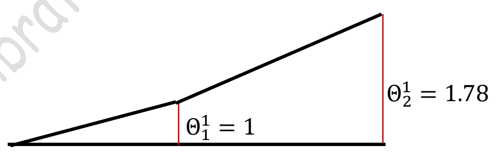
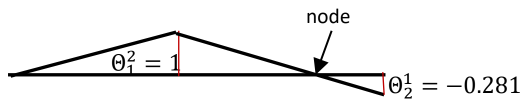
$$\begin{bmatrix} 20000 - 2 \times 106.8^2 & -10000 \\ -10000 & 10000 - 4 \times 106.8^2 \end{bmatrix} \begin{Bmatrix} \Theta_1^2 \\ \Theta_2^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$-2812.48\ \Theta_1^2 - 10000\ \Theta_2^2 = 0 \quad \Rightarrow\Rightarrow \quad \frac{\Theta_2^2}{\Theta_1^2} = -0.281$$

and $-10000\ \Theta_1^2 - 35624.96\ \Theta_2^2 = 0 \Rightarrow\Rightarrow \quad \frac{\Theta_2^1}{\Theta_1^1} = -0.281$

$$r_2 = -0.281$$

If $\Theta_1^2 = 1$, then $\Theta_2^1 = -0.281$

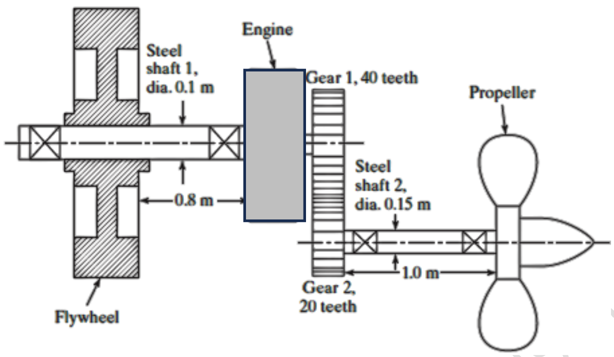
I mode shape	II mode shape
	
Normal modes or modal vectors $\{\Theta\}^1 = \begin{Bmatrix} {\Theta_1}^1 \\ {\Theta_2}^1 \end{Bmatrix} = {\Theta_1}^1 \begin{Bmatrix} 1 \\ 1.78 \end{Bmatrix}$	Normal modes or modal vectors $\{\Theta\}^2 = \begin{Bmatrix} {\Theta_1}^2 \\ {\Theta_2}^2 \end{Bmatrix} = {\Theta_1}^2 \begin{Bmatrix} 1 \\ -0.281 \end{Bmatrix}$

Example 3

The figure shows the schematic of an engine coupled to a proppellor through a pair of gears. The mass moment of inertia of the flywheel is very high compared to that of engine rotating parts, gears, and the propellor. Find the natural frequencies and mode shapes of the system approximating it as a two-rotor system undergoing torsional vibration.

Take $I_{engine} = 1000\ kg.m^2$, $I_{gear1} = 250\ kg.m^2$, $I_{gear2} = 150\ kg.m^2$, $I_{propellor} = 2000\ kg.m^2$

$$G_{steel} = 80\ GPa$$



Since the flywheel has a very large mass oment of inertia, the system can be assumed to be fixed at the flywheel. The engine and gear 1 are close to each other, the stiffness of the shaft between them can be neglected. To reduce the given system to a two rotor system, we need to convert it in to an equivalent system. The equivalent inertia can be found out by comparing the kinetic energies and equivalent stiffness can be found out by comparing the strain energies.

The number of teeth on the gear 2 (pinion) is twice the number of teeth on gear 1 (gear), the speed of the propellor and gear 2 $(\dot{\theta}_2)$will be twice the speed of engine and gear 1 $(\dot{\theta}_1)$.

$\dot{\theta}_2 = 2\dot{\theta}_1 = n\dot{\theta}_1$ where n is the speed ratio (driven speed/driver speed)

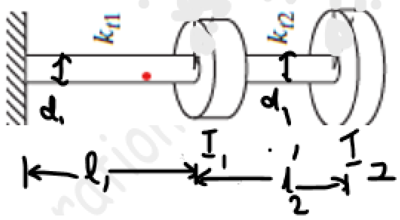
The kinetic energy of engine and gear 1, $T_1 = \frac{1}{2} I_e\,\dot{\theta}_1^{\ 2} + \frac{1}{2} I_{g1}\,\dot{\theta}_1^{\ 2} = \frac{1}{2}(I_e + I_{g1})\,\dot{\theta}_1^{\ 2}$

The kinetic energy of gear 2 and propellor, $T_2 = \frac{1}{2} I_{g2}\,\dot{\theta}_2^{\ 2} + \frac{1}{2} I_p\,\dot{\theta}_2^{\ 2} = \frac{1}{2}(I_{g2} + I_p)\,n^2\dot{\theta}_1^{\ 2}$

Kinetic energy of the system,

$$T_{system} = \frac{1}{2}(I_e + I_{g1})\,\dot{\theta}_1^{\ 2} + \frac{1}{2}(I_{g2} + I_p)\,n^2\dot{\theta}_1^{\ 2} = \frac{1}{2}(I_e + I_{g1} + n^2 I_{g2})\dot{\theta}_1^{\ 2} + \frac{1}{2}(n^2 I_p)\,\dot{\theta}_1^{\ 2}$$

If we are approximating the system as a two rotor system with engine speed $\dot{\theta}_1$, the engine, gear 1, and gear 2 are approximated as rotor 1 with inertia I_1 and propellor is replaced by another rotor 2 with inertia I_2 as shown.



The kinetic energy of the equivalent system,

$$T_{eq} = \frac{1}{2} I_1\,\dot{\theta}_1^{\ 2} + \frac{1}{2} I_2\,\dot{\theta}_1^{\ 2}$$

For both systems to be equivalent, the kinetic energy of both should be the same.

$$T_{eq} = T_{system}$$

$$\frac{1}{2} I_1\,\dot{\theta}_1^{\ 2} + \frac{1}{2} I_2\,\dot{\theta}_1^{\ 2} = \frac{1}{2}(I_e + I_{g1} + n^2 I_{g2})\dot{\theta}_1^{\ 2} + \frac{1}{2}(n^2 I_p)\dot{\theta}_1^{\ 2}$$

Comparing the terms on the right and left sides,

$$\boldsymbol{I_1 = I_e + I_{g1} + n^2 I_{g2} = 1000 + 250 + 2^2 \times 150 = 1850\ kg.m^2}$$

$$\boldsymbol{I_2 = n^2 I_p = 2^2 \times 2000 = 8000\ kg.m^2}$$

The torsional stiffness of the shaft 1 with diameter $d_1 = 0.1\ m$ and length $l_1 = 0.8\ m$ is equal to $k_1 = \frac{GJ_1}{l_1} = \frac{\left(80\times10^9 \frac{N}{m^2}\right)\times\left(\frac{\pi}{32}\times0.1^4 m^4\right)}{0.8\ m}$ where J_1 is the polar moment of inertia of the cross section of the shaft 1.

$$k_1 = 9.82 \times 10^5\ Nm/rad$$

The torsional stiffness of the shaft 2 with diameter $d_2 = 0.15\ m$ and length $l_2 = 1\ m$ is equal to $k_2 = \frac{GJ_2}{l_2} = \frac{\left(80\times10^9 \frac{N}{m^2}\right)\times\left(\frac{\pi}{32}\times0.15^4 m^4\right)}{1\ m}$ where J_2 is the polar moment of inertia of the cross section of the shaft 2.

$$k_2 = 39.8 \times 10^5\ Nm/rad$$

If θ_1 is the angle turned by shaft 1, then the angle turned by shaft 2 will be $\theta_2 = n\theta_1$

The strain energy of shaft 1, $U_1 = \frac{1}{2}k_1{\theta_1}^2$

The strain energy of shaft 2, $U_2 = \frac{1}{2}k_2{\theta_2}^2$

Total strain energy of the given system, $U_{system} = U_1 + U_2 = \frac{1}{2}k_1{\theta_1}^2 + \frac{1}{2}k_2{\theta_2}^2$

Assuming that the diameter and length of the shaft from fixed end to rotor 1 in the equivalent system is the same as that between the flywheel and engine $d_1 and\ l_1$, we can take $k_{t1} = k_1$. We can also assume that the shaft diameter from rotor 1 to rotor 2 is the same as d_1 (then length will be l_2').The stiffness of shaft from rotor 1 to rotor 2 is taken as k_{t2}

$$U_{eq} = \frac{1}{2}k_{t1}{\theta_1}^2 + \frac{1}{2}k_{t2}{\theta_2}^2$$

$$U_{eq} = U_{system}$$

$$\frac{1}{2}k_{t1}{\theta_1}^2 + \frac{1}{2}k_{t2}{\theta_2}^2 = \frac{1}{2}k_1{\theta_1}^2 + \frac{1}{2}k_2{\theta_2}^2$$

Comparing the terms on both sides

$$k_{t2} = k_2$$

The equation of motion for the first rotor

$$I_1\ddot{\theta}_1 = \Sigma M_{in\ the\ direction\ of\ \ddot{\theta}_1}$$

$$= k_2(\theta_2 - \theta_1) - k_1\theta_1$$

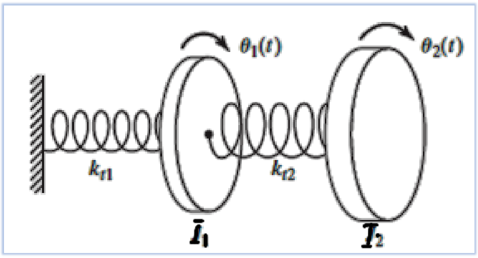
$$I_1\ddot{\theta}_1 + (k_1 + k_2)\theta_1 - k_2\theta_2 = 0 - - - - -(1)$$

The equation of motion for the second rotor

$$I_2\ddot{\theta}_2 = \Sigma M_{in\ the\ direction\ of\ \ddot{\theta}_2}$$

$$= -k_2(\theta_2 - \theta_1)$$

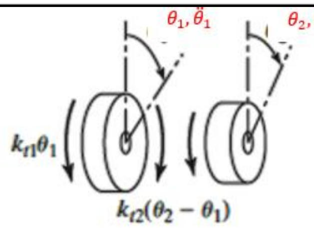
$$I_2\ddot{\theta}_2 - k_2\theta_1 + k_2\theta_2 = 0 - - - - -(2)$$



Combining the equations (1) and (2) in the matrix form

$$\begin{bmatrix} I_1 & 0 \\ 0 & I_2 \end{bmatrix}\begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}\begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} ----(3)$$

$$[I]\{\ddot{\theta}\} + [k]\{\theta\} = \{0\} ------(4)$$



Free body diagram

$$\begin{bmatrix} 1850 & 0 \\ 0 & 8000 \end{bmatrix}\begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} 9.82\times10^5 + 39.8\times10^5 & -39.8\times10^5 \\ -39.8\times10^5 & 39.8\times10^5 \end{bmatrix}\begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Assuming harmonic solution $\{\theta\}=\{\Theta\}\sin(\omega t+\phi) ---(4)$

where $\{\theta\}$ =angular displacements of rotors 1 and 2 $= \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix}$

$\{\Theta\}$ =amplitudes of angular displacements of rotors 1 and 2 $= \begin{Bmatrix} \Theta_1 \\ \Theta_2 \end{Bmatrix}$

Differentiating with respect to time, $\{\ddot{\theta}\} = -\omega^2\{\Theta\}\sin(\omega t+\phi)$

Eq. (4) becomes $-[I]\omega^2\{\Theta\}\sin(\omega t+\phi) + [k]\{\Theta\}\sin(\omega t+\phi) = \{0\}$

$$\{-\omega^2[I] + [k]\}\{\Theta\}\sin(\omega t+\phi) = \{0\} ---(5)$$

$$\{-\omega^2[I] + [k]\}\{\Theta\} = \{0\}$$

$$\left\{-\omega^2\begin{bmatrix} I_1 & 0 \\ 0 & I_2 \end{bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}\right\}\begin{Bmatrix} \Theta_1 \\ \Theta_2 \end{Bmatrix} = \{0\}$$

$$\begin{bmatrix} k_1+k_2-\omega^2 I_1 & -k_2 \\ -k_2 & k_2-\omega^2 I_2 \end{bmatrix}\begin{Bmatrix} \Theta_1 \\ \Theta_2 \end{Bmatrix} = \{0\} ----(6)$$

The condition for the system of linear homogeneous equations to have a non-trivial solutions, determinanat of the coefficient matrix=0

$$\begin{vmatrix} k_1+k_2-\omega^2 I_1 & -k_2 \\ -k_2 & k_2-\omega^2 I_2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 49.6\times10^5 - 1850\omega^2 & -39.8\times10^5 \\ -39.8\times10^5 & 39.8\times10^5 - 8000\omega^2 \end{vmatrix} = 0$$

$$(49.6\times10^5 - 1850\omega^2)(39.8\times10^5 - 8000\omega^2) - 15.84\times10^{12} = 0$$

$$19.74\times10^{12} - 396\times10^8\omega^2 - 73.63\times10^8\omega^2 + 1.48\times10^7\omega^4 - 15.84\times10^{12} = 0$$

$$1.48\times10^7\omega^4 - 469.63\times10^8\omega^2 + 390.8\times10^{10} = 0$$

Substituting $\omega^2 = \lambda$

$$1.48\times10^7\lambda^2 - 469.63\times10^8\lambda + 390.8\times10^{10} = 0$$

$$\lambda = \frac{469.63\times10^8 \pm \sqrt{2205.5\times10^{18} - 231.4\times10^{18}}}{2\times1.48\times10^7} = \frac{469.63\times10^8 \pm 444.3\times10^8}{0.296\times10^8}$$

$$= 85.57 \ and \ 3087.6$$

$$\therefore \omega = 9.25\frac{rad}{s};\ 55.6\frac{rad}{s}$$

The natural frequecies of the two rotor system are $\omega_1 = 9.25\,rad/s$ and $\omega_2 = 55.6\,rad/s$

When $\omega = \omega_1 = 9.25 rad/s$ (First Mode of vibration), the equation (6) becomes

$$\begin{bmatrix} k_1+k_2-\omega_1{}^2 I_1 & -k_2 \\ -k_2 & k_2-\omega_1{}^2 I_2 \end{bmatrix}\begin{Bmatrix} \Theta_1^1 \\ \Theta_2^1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} 49.6\times10^5 - 1850\times9.25^2 & -39.8\times10^5 \\ -39.8\times10^5 & 39.8\times10^5 - 8000\times9.25^2 \end{bmatrix}\begin{Bmatrix} \Theta_1^1 \\ \Theta_2^1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$4801709\ \Theta_1^1 - 39.8 \times 10^5\ \Theta_2^1 = 0 \quad \Rightarrow\Rightarrow \quad \frac{\Theta_2^1}{\Theta_1^1} = 1.21$$

$$\text{and } -39.8 \times 10^5\ \Theta_1^1 + 3295500\ \Theta_2^1 = 0 \Rightarrow\Rightarrow \quad \frac{\Theta_2^1}{\Theta_1^1} = 1.21$$

$$r_1 = 1.21$$

$$\text{If } \Theta_1^1 = 1, \text{ then } \Theta_2^1 = 1.21$$

When $\omega = \omega_2 = 55.6\, rad/s$ (Second Mode of vibration), the equation (6) becomes

$$\begin{bmatrix} k_1 + k_2 - {\omega_2}^2 I_1 & -k_2 \\ -k_2 & k_2 - {\omega_2}^2 I_2 \end{bmatrix} \begin{Bmatrix} \Theta_1^2 \\ \Theta_2^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

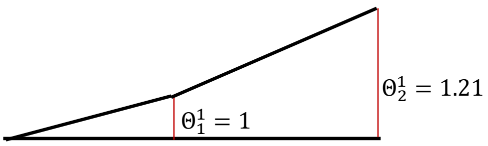
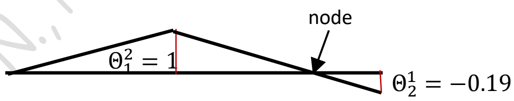
$$\begin{bmatrix} 49.6 \times 10^5 - 1850 \times 55.6^2 & -39.8 \times 10^5 \\ -39.8 \times 10^5 & 39.8 \times 10^5 - 8000 \times 55.6^2 \end{bmatrix} \begin{Bmatrix} \Theta_1^2 \\ \Theta_2^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$-759016\ \Theta_1^2 - 39.8 \times 10^5\ \Theta_2^2 = 0 \quad \Rightarrow\Rightarrow \quad \frac{\Theta_2^2}{\Theta_1^2} = -0.19$$

$$\text{and } -39.8 \times 10^5\ \Theta_1^2 - 20750880\ \Theta_2^2 = 0 \Rightarrow\Rightarrow \quad \frac{\Theta_2^1}{\Theta_1^1} = -0.19$$

$$r_2 = -0.19$$

$$\text{If } \Theta_1^2 = 1, \text{ then } \Theta_2^1 = -0.19$$

I mode shape	II mode shape
	
Normal modes or modal vectors	Normal modes or modal vectors
$\{\Theta\}^1 = \begin{Bmatrix} {\Theta_1}^1 \\ {\Theta_2}^1 \end{Bmatrix} = {\Theta_1}^1 \begin{Bmatrix} 1 \\ 1.21 \end{Bmatrix}$	$\{\Theta\}^2 = \begin{Bmatrix} {\Theta_1}^2 \\ {\Theta_2}^2 \end{Bmatrix} = {\Theta_1}^2 \begin{Bmatrix} 1 \\ -0.19 \end{Bmatrix}$

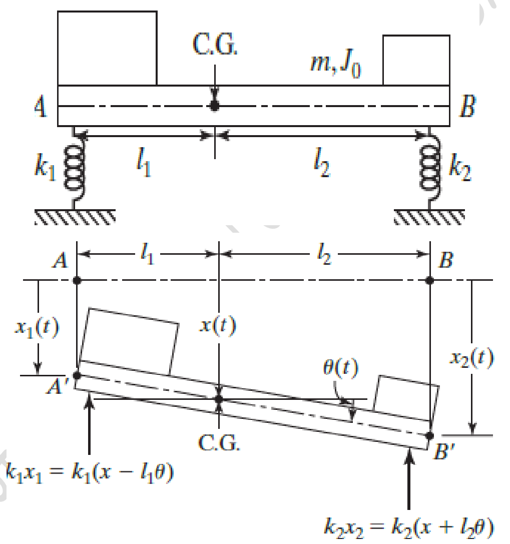
Coordinate Coupling and Principal Coordinates

From the definition, it is understood that an n-degree of freedom system will have n independent coordinates to decribe its configuation completely. These coordinates are usually the geometrical properties measured from the equilibrium position of the body which is vibrating. However, it is possible to select some other set of n-coordinates to describe the configuration of the system. Each of these sets of n coordinates is called the generalised coordinates.

Consider the model of the lathe or the car given in the beginning of this chapter. Take m as the mass of the lathe and J_o as the mass moment of inertia of the lathe with respect to the axis through its CG. Any of the following sets of coordinates can be used to describe the motion.

i) Deflection $x(t)$ of the C.G and rotation $\theta(t)$
ii) Deflections $x_1(t)$ and $x_2(t)$ of the two ends
iii) Deflection $x_1(t)$ of the end and rotation $\theta(t)$
iv) Deflection $y(t)$ of any point P and rotation $\theta(t)$

Any set of these coordinates (x, θ), (x_1, x_2), (x_1, θ), or (y, θ) represents the generalised coordinates of the system.



Equations of motion using coordinates (x, θ)::

$m\ddot{x} = \Sigma\,(forces\ in\ the\ direction\ of\ \ddot{x})$ and

$J_o\ddot{\theta} = \Sigma\,(Moments\ of\ forces\ about\ the\ axis\ through\ CG\ in\ the\ direction\ of\ \ddot{\theta})$

$$m\ddot{x} = -k_1(x - l_1\theta) - k_2(x + l_2\theta)$$

$$J_o\ddot{\theta} = k_1(x - l_1\theta)l_1 - k_2(x + l_2\theta)l_2$$

$$m\ddot{x} + k_1x - k_1l_1\theta + k_2x + k_2l_2\theta = 0$$

$$\boldsymbol{m\ddot{x} + (k_1 + k_2)x + (k_2l_2 - k_1l_1)\theta = 0} \dots\dots \mathbf{(1)}$$

$$J_o\ddot{\theta} - k_1l_1(x - l_1\theta) + k_2l_2(x + l_2\theta) = 0$$

$$\boldsymbol{J_o\ddot{\theta} + (k_2l_2 - k_1l_1)x + (k_1{l_1}^2 + k_2{l_2}^2)\theta = 0} \dots . \mathbf{(2)}$$

From Eqns (1) and (2) above

$$\begin{bmatrix} m & 0 \\ 0 & J_o \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & (k_2l_2 - k_1l_1) \\ (k_2l_2 - k_1l_1) & (k_1{l_1}^2 + k_2{l_2}^2) \end{bmatrix} \begin{Bmatrix} \boldsymbol{x} \\ \boldsymbol{\theta} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \dots\dots\dots (3)$$

It is to be noted that both the equations (1) and (2) contains the coordinates (x, θ)

The mass or inertia matrix is diagonal but the stiffness/elastic matrix is non diagonal. If the inertia matix is non-diagonal, it is called dynamic coupling or inertia coupling. If the stiffness matrix is non-diagonal, it is known as elastic coupling or static coupling.

If $k_2l_2 = k_1l_1$, the stiffness matrix becomes diagonal (or de-coupled). Then these equations will be two independent second order differetial equations in x and θ. If the equations are coupled, any initial conditions on x or θ will initiate both translational and rotational motions. But, if the equations are un-coupled, any initial conditions on x will initiate only tranlatory motion and any initial conditions on θ will produce only rotational oscillations.

Equations of motion using coordinates (y, θ)::

From the figure,

$$x = y + e\theta \qquad \Rightarrow \qquad \ddot{x} = \ddot{y} + e\ddot{\theta}$$

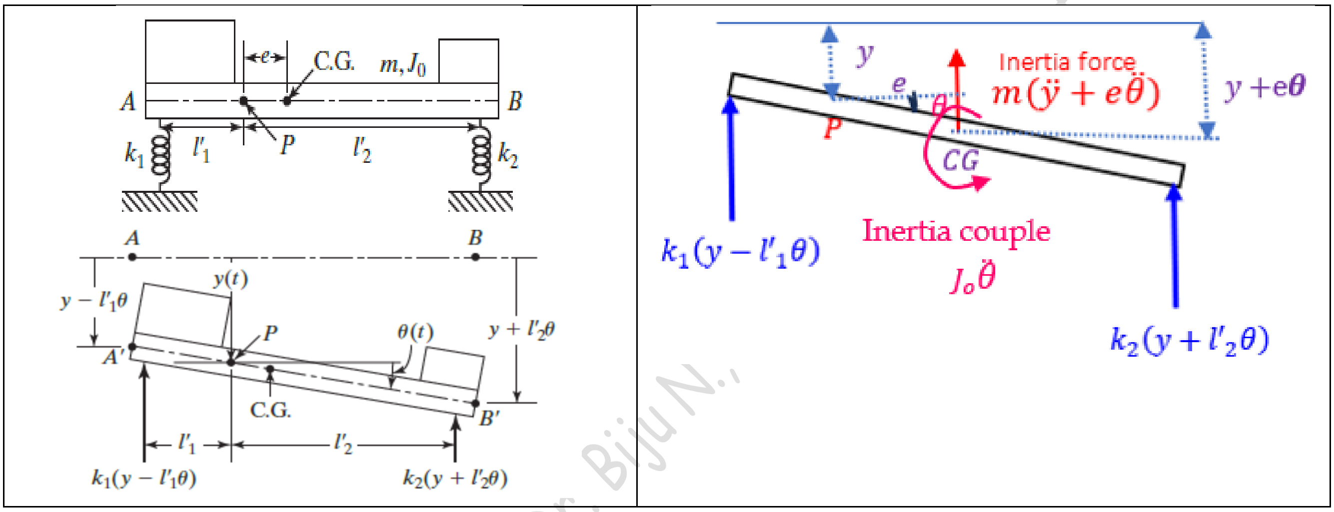
Using D'Alembert's Principle

Σ (all $forces$ $including$ the $inertia$ $force$) $= 0$ and

Σ ($Moments$ of $forces$ $including$ the $inertia$ $forces$ $about$ the $axis$ $through$ any $point$ P) $= 0$

$$m(\ddot{y} + e\ddot{\theta}) + k_1(y - l'_1\theta) + k_2(y + l'_2\theta) = 0$$

$$J_o\ddot{\theta} - k_1(y - l'_1\theta)l'_1 + k_2(y + l'_2\theta)l'_2 + m(\ddot{y} + e\ddot{\theta})e = 0$$



$$m\ddot{y} + me\ddot{\theta} + (k_1 + k_2)y + (k_2l'_2 - k_1l'_1)\theta = 0$$

$$me\ddot{y} + (me^2 + J_o)\ddot{\theta} + (k_2l'_2 - k_1l'_1)y + (k_1{l'_1}^2 + k_2{l'_2}^2)\theta = 0$$

$$\begin{bmatrix} m & me \\ me & me^2 + J_o \end{bmatrix}\begin{Bmatrix} \ddot{y} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & (k_2l'_2 - k_1l'_1) \\ (k_2l'_2 - k_1l'_1) & (k_1{l'_1}^2 + k_2{l'_2}^2) \end{bmatrix}\begin{Bmatrix} \boldsymbol{y} \\ \boldsymbol{\theta} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

We know from parallel axis theorem,

$$me^2 + J_o = J_P$$

$$\begin{bmatrix} m & me \\ me & J_P \end{bmatrix}\begin{Bmatrix} \ddot{y} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & (k_2l'_2 - k_1l'_1) \\ (k_2l'_2 - k_1l'_1) & (k_1{l'_1}^2 + k_2{l'_2}^2) \end{bmatrix}\begin{Bmatrix} \boldsymbol{y} \\ \boldsymbol{\theta} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \dots\dots\dots (4)$$

Here both the inertia matrix and stiffness matrix are non-diagonal. It is statically and dynamically coupled. If $k_2l'_2 = k_1l'_1$, it becomes only dynamically coupled.

- In the most general case, a viscously damped two degree of freedom system has equations of motion as

$$\begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Note that the mass, damping, and stiffness matrices will be symmetric.

$$m_{21} = m_{12}; \quad c_{21} = c_{12}\,; \quad k_{21} = k_{12}$$

If the mass matrix is not diagonal, the system has inertia coupling

If the damping matrix is not diagonal, the system has damping or velocity coupling.

If the stiffness matrix is not diagonal, the system has static coupling.

Both inertia coupling and velocity coupling comes under dynamic coupling.

- The system vibrates on its own natural way regardless of the coordinates chosen. The choice of coordinates is only a convenience.
- The coupling is not an inherent property of the system, instead it depends on the chosen coordinates.
- It is possible to choose a coordinate system q_1 and q_2 which will give the equation of motion statically and dynamically uncoupled. Such coordinates are called Principal Coordinates.
- The advantage of principal coordinates are that the uncoupled equations of motion can be solved independently of each other.

For example, consider the solution of the Example 1

$$x_1(t) = 0.15\cos 44.72t - 0.05\cos 76.46t$$
$$x_2(t) = 0.15\cos 44.72t + 0.05\cos 76.46t$$

Suppose we choose two coordinates

$$q_1(t) = 0.15\cos 44.72t$$
$$q_2(t) = 0.05\cos 76.46t$$

These are the solutions of the differential equations

$$\ddot{q}_1 + 44.72^2 q_1 = 0$$
$$\ddot{q}_2 + 76.46^2 q_2 = 0$$

[The solution of a differential equation $\ddot{x} + \omega^2 x = 0$ is $x = A\sin(\omega t + \varphi)$]

These equation can be written in matrix form

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{Bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{Bmatrix} + \begin{bmatrix} 44.72^2 & 0 \\ 0 & 76.46^2 \end{bmatrix}\begin{Bmatrix} q_1 \\ q_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Since these two equations are statically and dynamically not coupled, $q_1(t)$and $q_2(t)$ are principal coordinates. Now the solution can be written in terms of the principal coordinates

$$x_1(t) = q_1(t) - q_2(t)$$
$$x_2(t) = q_1(t) + q_2(t)$$

From this, we can find out the principal coordinates

$$q_1(t) = \frac{x_2(t) + x_1(t)}{2}$$
$$q_2(t) = \frac{x_2(t) - x_1(t)}{2}$$

Example 4

Determine the pitch (angular motion) and bounce (up-down motion) frequencies and the location of nodes (oscillation centres) of a car with the following data

Mass (m)=1000 kg

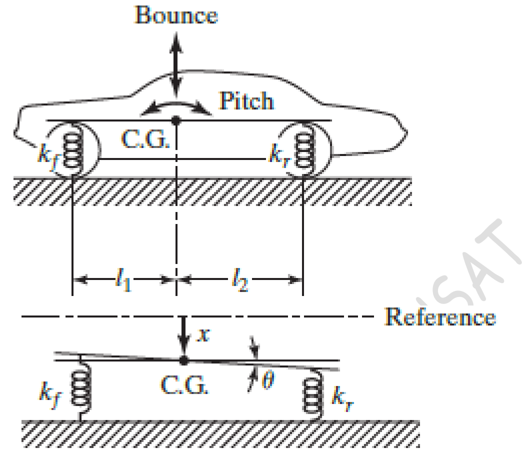
Radius of gyration about CG (r)=0.9 m

Distance between front axle and CG (l_1)= 1 m

Distance between rear axle and CG (l_2)= 1.5 m

Front spring stiffness (k_1)= 18 kN/m

Rear spring stiffness (k_2) = 22 kN/m



The mass moment of inertia of the car with respect to the centroidal axis

$$J_G = mr^2 = 1000 \times 0.9^2 = 810\ kg.m^2$$

Choosing the displacement $x(t)$ of the CG and rotaion $\theta(t)$ about the axis through CG as the coordinates, the equtions of motion can be written as

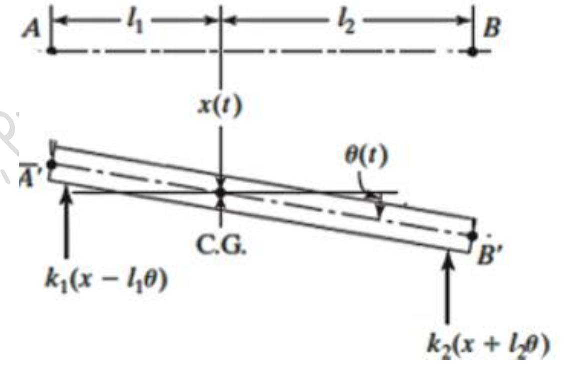
$$m\ddot{x} = -k_1(x - l_1\theta) - k_2(x + l_2\theta)$$

$$J_G\ddot{\theta} = k_1(x - l_1\theta)l_1 - k_2(x + l_2\theta)l_2$$

Or

$$m\ddot{x} + (k_1 + k_2)x + (k_2l_2 - k_1l_1)\theta = 0$$

$$J_G\ddot{\theta} + (k_2l_2 - k_1l_1)x + (k_1{l_1}^2 + k_2{l_2}^2)\theta = 0$$



From the above equations

$$\begin{bmatrix} m & 0 \\ 0 & J_G \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & (k_2l_2 - k_1l_1) \\ (k_2l_2 - k_1l_1) & (k_1{l_1}^2 + k_2{l_2}^2) \end{bmatrix} \begin{Bmatrix} \boldsymbol{x} \\ \boldsymbol{\theta} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} 1000 & 0 \\ 0 & 810 \end{bmatrix} \begin{Bmatrix} \ddot{x} \\ \ddot{\theta} \end{Bmatrix} + \begin{bmatrix} 40000 & 15000 \\ 15000 & 67500 \end{bmatrix} \begin{Bmatrix} \boldsymbol{x} \\ \boldsymbol{\theta} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$[M]\{\ddot{\Omega}\} + [K]\{\Omega\} = \{0\} \text{ where } \{\Omega\} = \begin{Bmatrix} x \\ \theta \end{Bmatrix}$$

Assuming harmonic solution $\{\Omega\}=\{\Psi\}\sin(\omega t + \phi)$ where $\{\Psi\} = \begin{Bmatrix} X \\ \Theta \end{Bmatrix}$

Differentiating with respect to time, $\{\ddot{\Omega}\} = -\omega^2\{\Psi\}\sin(\omega t + \phi)$

Eq. (4) becomes $-[M]\omega^2\{\Psi\}\sin(\omega t + \phi) + [K]\{\Psi\}\sin(\omega t + \phi) = \{0\}$

$$\{-\omega^2[M] + [K]\}\{\Psi\}\sin(\omega t + \phi) = \{0\} - - - (5)$$

$$\{-\omega^2[M] + [K]\}\{\Psi\} = \{0\}$$

$$\left\{-\omega^2 \begin{bmatrix} 1000 & 0 \\ 0 & 810 \end{bmatrix} + \begin{bmatrix} 40000 & 15000 \\ 15000 & 67500 \end{bmatrix}\right\} \begin{Bmatrix} X \\ \Theta \end{Bmatrix} = \{0\}$$

$$\begin{bmatrix} 40000 - \omega^2 1000 & 15000 \\ 15000 & 67000 - \omega^2 810 \end{bmatrix} \begin{Bmatrix} X \\ \Theta \end{Bmatrix} = \{0\}$$

The condition for the system of linear homogeneous equations to have a non-trivial solutions, determinanat of the coefficient matrix = 0

$$\begin{vmatrix} 40000-\omega^2 1000 & 15000 \\ 15000 & 67000-\omega^2 810 \end{vmatrix} = 0$$

$$(40000-1000\omega^2)(\,67000-810\omega^2)-15000\times 15000=0$$

$$26.8\times 10^8-32.4\times 10^6\omega^2-67\times 10^6\omega^2+81\times 10^4\omega^4-2.25\times 10^8=0$$

$$81\times 10^4\omega^4-99.4\times 10^6\omega^2+24.55\times 10^8=0$$

Substituting $\omega^2=\lambda$

$$81\times 10^4\lambda^2-99.4\times 10^6\,\lambda+24.55\times 10^8=0$$

$$\lambda=\frac{99.4\times 10^6\pm\sqrt{9880.36\times 10^{12}-7954.2\times 10^{12}}}{2\times 81\times 10^4}=\frac{99.4\times 10^6\pm 43.89\times 10^6}{162\times 10^4}$$

$$=\;34.3\;and\;88.5$$

$$\therefore\;\omega=5.85\,rad/s;\;\;9.41\;\;rad/s$$

The natural frequecies of the system are $\omega_1=5.85\,rad/s$ and $\omega_2=9.41\,rad/s$

When $\omega=\omega_1=5.85\,rad/s$ (First Mode of vibration), the equation (6) becomes

$$\begin{bmatrix} 40000-1000{\omega_1}^2 & 15000 \\ 15000 & 67000-810{\omega_1}^2 \end{bmatrix}\begin{Bmatrix} X^1 \\ \Theta^1 \end{Bmatrix}=\{0\}$$

$$\begin{bmatrix} 40000-1000\times 5.85^2 & 15000 \\ 15000 & 67000-810\times 5.85^2 \end{bmatrix}\begin{Bmatrix} X^1 \\ \Theta^1 \end{Bmatrix}=\{0\}$$

$$\begin{bmatrix} 5777.5 & 15000 \\ 15000 & 39280 \end{bmatrix}\begin{Bmatrix} X^1 \\ \Theta^1 \end{Bmatrix}=\begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$5777.5X^1+15000\;\Theta^1=0\qquad\Rightarrow\Rightarrow\qquad\frac{X^1}{\Theta^1}=-2.6461$$

$$r_1=-2.6461$$

When $\omega=\omega_1=9.41\,rad/s$ (second Mode of vibration), the equation becomes

$$\begin{bmatrix} 40000-1000{\omega_2}^2 & 15000 \\ 15000 & 67000-810{\omega_2}^2 \end{bmatrix}\begin{Bmatrix} X^2 \\ \Theta^2 \end{Bmatrix}=\{0\}$$

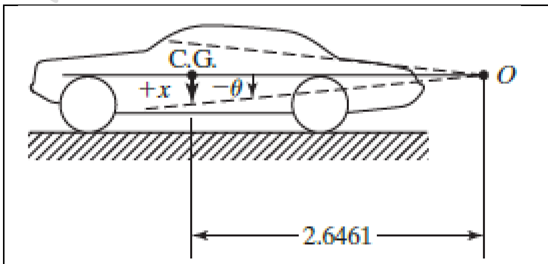
$$\begin{bmatrix} 40000-1000\times 9.41^2 & 15000 \\ 15000 & 67000-810\times 9.41^2 \end{bmatrix}\begin{Bmatrix} X^2 \\ \Theta^2 \end{Bmatrix}=\{0\}$$

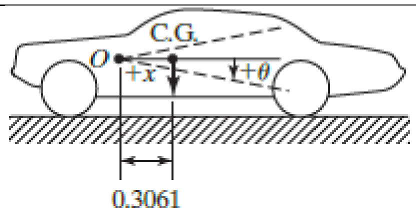
$$\begin{bmatrix} -48548.1 & 15000 \\ 15000 & -4723.9 \end{bmatrix}\begin{Bmatrix} X^2 \\ \Theta^2 \end{Bmatrix}=\begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$-48548.1X^2+15000\;\Theta^2=0\qquad\Rightarrow\Rightarrow\qquad\frac{X^2}{\Theta^2}=0.3061$$

$$r_2=0.3061$$

The location of node





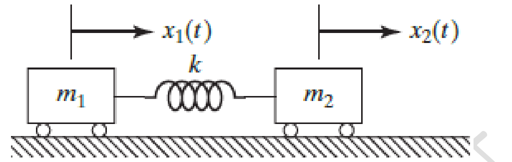
$$\tan\Theta^2=\frac{X^2}{OG}\Longrightarrow OG=\frac{X^2}{\tan\Theta^2}\simeq\frac{X^2}{\Theta^2}=0.3061$$

$$\tan\Theta^1 = \frac{X^1}{OG} \Longrightarrow OG = \frac{X^1}{\tan\Theta^1} \backsimeq \frac{X^1}{\Theta^1} = 2.6461$$

Example 5

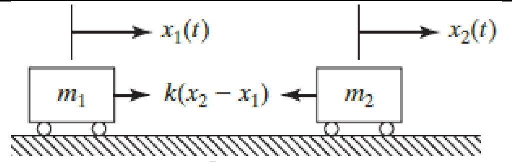
The figure shows the model of two railway coaches connected by a coupling. Find the natural frequencies and modeshapes of the system by taking typical values $m_1 = 1\,Kg, m_2 = 2\,Kg, k = 200N/m,$

$$x_1(0) = 0.1m,\; x_2(0) = \dot{x}_1(0) = \dot{x}_2(0) = 0$$



The equations of motion are

$m_1\ddot{x}_1 = k(x_2 - x_1)$ and $m_2\ddot{x}_2 = -k(x_2 - x_1)$



$$m_1\ddot{x}_1 + kx_1 - kx_2 = 0 \qquad\qquad m_2\ddot{x}_2 - kx_1 + kx_2 = 0$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} 200 & -200 \\ -200 & 200 \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

This system of linear homogeneous equations can be written in matrix form as

$$[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$$

Assuming a harmonic solution

$$x_1(t) = X_1 \sin(\omega t + \phi)$$
$$x_2(t) = X_2 \sin(\omega t + \phi)$$

$x_1(t)$ and $x_2(t)$ are the displacements of mass m_1 and m_2 at any time

X_1 and X_2 are the amplitudes of masses, ω the frequency, and ϕ the phase angle

Combining the solutions

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} \sin(\omega t + \phi)$$

$$\{x\}=\{X\}\sin(\omega t + \phi)$$

Differentiating the above equation two times, we get

$$\{\ddot{x}\} = -\omega^2\{X\}\sin(\omega t + \phi)$$

Therefore, the matrix equation now becomes

$$-\omega^2[M]\{X\}\sin(\omega t + \phi) + [K]\{X\}\sin(\omega t + \phi) = \{0\}$$
$$([K] - \omega^2[M])\{X\}\sin(\omega t + \phi) = \{0\}$$
$$([K] - \omega^2[M])\{X\} = \{0\} --- (1)$$

For this system of linear homogeneous equations to have a non-trivial solution,

$$\det\{[K] - \omega^2[M]\} = 0$$

$$\begin{vmatrix} 200 - \omega^2 & -200 \\ -200 & 200 - 2\omega^2 \end{vmatrix} = 0$$

$$(200 - \omega^2)(200 - 2\omega^2) - 40000 = 0$$
$$40000 - 400\omega^2 - 200\omega^2 + 2\omega^4 - 40000 = 0$$
$$\omega^2(2\omega^2 - 600) = 0$$

$$\omega^2 = 0 \; or \; (2\omega^2 - 600) = 0$$

$$\omega = 0, 17.32 \; rad/s$$

The natural frequecies of the system are $\omega_1 = 0\,rad/s$ and $\omega_2 = 17.32\,rad/s$. One of the natural frequencies of the system is zero. The system will have a rigid body motion (a traslation in this case) and a vibration. Such systems are known as semi definite systems, unrestrained systems, or degenerate systems

When $\omega = \omega_1 = 0$ (First Mode of vibration), the equation (1) becomes

$$\begin{bmatrix} 200 - {\omega_1}^2 & -200 \\ -200 & 200 - 2{\omega_1}^2 \end{bmatrix} \begin{Bmatrix} X_1^1 \\ X_2^1 \end{Bmatrix} = \{0\}$$

$$\begin{bmatrix} 200 & -200 \\ -200 & 200 \end{bmatrix} \begin{Bmatrix} X_1^1 \\ X_2^1 \end{Bmatrix} = \{0\}$$

$$200X_1^1 - 200\,X_2^1 = 0 \quad \Rightarrow\Rightarrow \quad \frac{X_2^1}{X_1^1} = 1$$

$$r_1 = 1$$

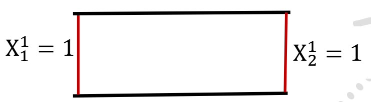
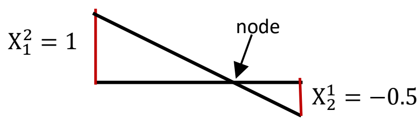
When $\omega = \omega_1 = 17.32\,rad/s$ (First Mode of vibration), the equation (1) becomes

$$\begin{bmatrix} 200 - {\omega_2}^2 & -200 \\ -200 & 200 - 2{\omega_2}^2 \end{bmatrix} \begin{Bmatrix} X_1^2 \\ X_2^2 \end{Bmatrix} = \{0\}$$

$$\begin{bmatrix} 200 - 17.32^2 & -200 \\ -200 & 200 - 2 \times 17.32^2 \end{bmatrix} \begin{Bmatrix} X_1^2 \\ X_2^2 \end{Bmatrix} = \{0\}$$

$$-100X_1^2 - 200\,X_2^2 = 0 \quad \Rightarrow\Rightarrow \quad \frac{X_2^2}{X_1^2} = -0.5$$

$$r_2 = 0.5$$

I mode shape	II mode shape
	
Rigid body motion	
Normal modes or modal vectors	Normal modes or modal vectors
$\{x\}^1 = \begin{Bmatrix} x_1^1 \\ x_2^1 \end{Bmatrix} = x_1^1 \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$	$\{x\}^2 = \begin{Bmatrix} x_1^2 \\ x_2^2 \end{Bmatrix} = x_1^2 \begin{Bmatrix} 1 \\ -0.5 \end{Bmatrix}$

The resulting motion can be obtained by the linear combination (superposition) of the two normal modes (Refer Example 1). The general solution can be written as

$$\vec{x}(t) = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

$$= \vec{x}^1(t) + \vec{x}^2(t)$$

$$= \{\vec{X}^1\} \sin(\omega_1 t + \phi_1) + \{\vec{X}^2\} \sin(\omega_2 t + \phi_2)$$

$$= \begin{Bmatrix} X_1^1 \\ X_2^1 \end{Bmatrix} sin(\omega_1 t + \phi_1) + \begin{Bmatrix} X_1^2 \\ X_2^2 \end{Bmatrix} sin(\omega_2 t + \phi_2)$$

$$= \begin{Bmatrix} X_1^1 \\ r_1 X_1^1 \end{Bmatrix} \sin(\omega_1 t + \phi_1). + \begin{Bmatrix} X_1^2 \\ r_2 X_1^2 \end{Bmatrix} \sin(\omega_2 t + \phi_2).$$

$$= \begin{Bmatrix} X_1^1\; \sin(\omega_1 t + \phi_1) + X_1^2\; \sin(\omega_2 t + \phi_2) \\ r_1 X_1^1\; \sin(\omega_1 t + \phi_1) + r_2 X_1^2 \sin(\omega_2 t + \phi_2) \end{Bmatrix} - - - (2)$$

The initial displacement of the masses; $x_1(t = 0) = x_1(0)$ and $x_2(t = 0) = x_2(0)$

The initial velocity of the masses; $\dot{x}_1(t=0)=\dot{x}_1(0)$ and $\dot{x}_2(t=0)=\dot{x}_2(0)$

We have the displacement equations

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1^1 \sin(\omega_1 t+\phi_1) + X_1^2 \sin(\omega_2 t+\phi_2) \\ r_1 X_1^1 \sin(\omega_1 t+\phi_1) + r_2 X_1^2 \sin(\omega_2 t+\phi_2) \end{Bmatrix} ----(3)$$

Differentiating with respect to time

$$\begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{Bmatrix} = \begin{Bmatrix} \omega_1 X_1^1 \cos(\omega_1 t+\phi_1) + \omega_2 X_1^2 \cos(\omega_2 t+\phi_2) \\ \omega_1 r_1 X_1^1 \cos(\omega_1 t+\phi_1) + \omega_2 r_2 X_1^2 \cos(\omega_2 t+\phi_2) \end{Bmatrix} ----(4)$$

$$X_1^1 = \sqrt{\left\{\frac{x_2(0)-r_2x_1(0)}{(r_1-r_2)}\right\}^2 + \left\{\frac{\dot{x}_2(0)-\dot{x}_1(0)}{\omega_1(r_1-r_2)}\right\}^2} --(5)$$

$$\tan\phi_1 = \omega_1\left(\frac{x_2(0)-r_2x_1(0)}{\dot{x}_2(0)-\dot{x}_1(0)}\right) --(6)$$

$$X_1^2 = \sqrt{\left\{\frac{r_1\,x_1(0)-x_2(0)}{(r_1-r_2)}\right\}^2 + \left\{\dot{x}_1(0)-\left(\frac{\dot{x}_2(0)-\dot{x}_1(0)}{\omega_2(r_1-r_2)}\right)\right\}^2} --(7)$$

$$\tan\phi_2 = \omega_2\left(\frac{r_1\,x_1(0)-x_2(0)}{\omega_2\dot{x}_1(0)(r_1-r_2)-(\dot{x}_2(0)-\dot{x}_1(0))}\right) --(8)$$

Initial conditions given in the problem are

$$x_1(0)=0.1m; \qquad x_2(0)=0$$

$$\dot{x}_1(0)=0; \qquad \dot{x}_2(0)=0$$

$$\tan\phi_1 = \omega_1\left(\frac{x_2(0)-r_2x_1(0)}{\dot{x}_2(0)-\dot{x}_1(0)}\right) = 0 \Rightarrow \phi_1 = 0$$

$$\tan\phi_2 = \omega_2\left(\frac{r_1\,x_1(0)-x_2(0)}{\omega_2\dot{x}_1(0)(r_1-r_2)-(\dot{x}_2(0)-\dot{x}_1(0))}\right) = \infty \Rightarrow \phi_2 = \frac{\pi}{2}$$

Substituting the values of $\phi_1 = 0;\ \phi_2 = \frac{\pi}{2}$ in (5) and (7)

$$X_1^1 = \frac{x_2(0)-r_2x_1(0)}{(r_1-r_2)} = \frac{0-(-0.5)\times 0.1}{(1-(-0.5))} = 0.033$$

$$X_1^2 = \frac{r_1\,x_1(0)-x_2(0)}{(r_1-r_2)} = \frac{(1)\times(0.1)-0}{(1-(-0.5))} = -0.067$$

The general solution is

$$\begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} X_1^1 \sin(\omega_1 t+\phi_1) + X_1^2 \sin(\omega_2 t+\phi_2) \\ r_1 X_1^1 \sin(\omega_1 t+\phi_1) + r_2 X_1^2 \sin(\omega_2 t+\phi_2) \end{Bmatrix}$$

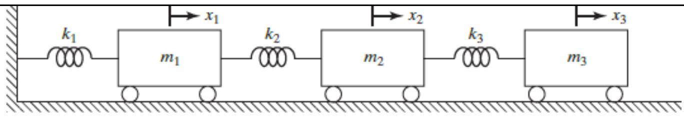
$$= \begin{Bmatrix} 0.033\ \sin(0\times t+\frac{\pi}{2}) - 0.067\ \sin(17.32t+\frac{\pi}{2}) \\ 1\times 0.033\ \sin(0\times t+\frac{\pi}{2}) + (-0.5)\times(-0.067)\sin(17.32t+\frac{\pi}{2}) \end{Bmatrix}$$

$$x_1 = 0.033 - 0.067\cos 17.32t$$

$$x_2 = 0.033 + 0.033\cos 17.32t$$

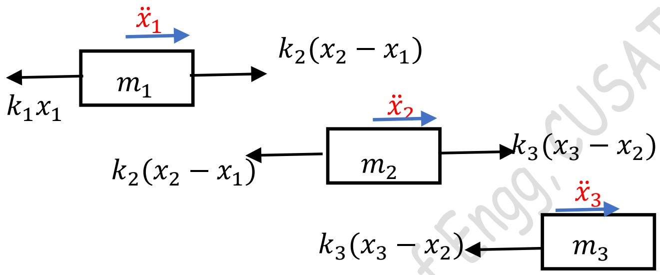
Example 6

Find the natural frequencies and modeshapes of the system if $m_1 = m_2 = m_3 = m;\ \ k_1 = k_2 = k_3 = k$



Equations of motion

$$m_1\ddot{x}_1 = k_2(x_2 - x_1) - k_1x_1$$
$$m_2\ddot{x}_2 = -k_2(x_2 - x_1) + k_3(x_3 - x_2)$$
$$m_3\ddot{x}_3 = -k_3(x_3 - x_2)$$



$$m_1\ddot{x}_1 + (k_1 + k_2)x_1 - k_2x_2 = 0$$
$$m_2\ddot{x}_2 - k_2x_1 + (k_2 + k_3)x_2 - k_3x_3 = 0$$
$$m_3\ddot{x}_3 - k_3x_2 + k_3x_3 = 0$$

These equations of motion can be written in matrix form

$$\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{Bmatrix} + \begin{bmatrix} (k_1 + k_2) & -k_2 & 0 \\ -k_2 & (k_2 + k_3) & -k_3 \\ 0 & -k_3 & k_3 \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix}\begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{Bmatrix} + \begin{bmatrix} 2k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & k \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

This system of linear homogeneous equations can be written in matrix form as

$$[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$$

Assuming a harmonic solution

$$x_1(t) = X_1 \sin(\omega t + \phi)$$
$$x_2(t) = X_2 \sin(\omega t + \phi)$$
$$x_3(t) = X_3 \sin(\omega t + \phi$$

This can be written as $\{x\}=\{X\}\sin(\omega t + \phi)$, where $\{x\} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}$ and $\{X\} = \begin{Bmatrix} X_1 \\ X_2 \\ X_3 \end{Bmatrix}$

Differentiating the above equation two times, we get

$$\{\ddot{x}\} = -\omega^2\{X\}\sin(\omega t + \phi)$$

Therefore, the matrix equation now becomes

$$-\omega^2[M]\{X\}\sin(\omega t + \phi) + [K]\{X\}\sin(\omega t + \phi) = \{0\}$$
$$([K] - \omega^2[M])\{X\}\sin(\omega t + \phi) = \{0\}$$
$$([K] - \omega^2[M])\{X\} = \{0\} - - - (1)$$

For this system of linear homogeneous equations to have a non-trivial solution,

$$\det\{[K] - \omega^2[M]\} = 0$$

$$\begin{vmatrix} 2k - \omega^2 m & -k & 0 \\ -k & 2k - \omega^2 m & -k \\ 0 & -k & k - \omega^2 m \end{vmatrix} = 0$$

$$(2k - \omega^2 m)(2k - \omega^2 m)(k - \omega^2 m) - k^2(2k - \omega^2 m) - k^2(k - \omega^2 m) = 0$$

$$(4k^2-4km\omega^2+m^2\omega^4)(k-\omega^2 m)-3k^3+2k^2m\omega^2=0$$

$$(4k^3-4k^2m\omega^2-4k^2m\omega^2+4km^2\omega^4+km^2\omega^4-m^3\omega^6)-3k^3+2k^2m\omega^2=0$$

$$m^3\omega^6-5km^2\omega^4+6k^2m\omega^2-k^3=0$$

Dividing by k^3

$$\left(\frac{m\omega^2}{k}\right)^3-5\left(\frac{m\omega^2}{k}\right)^2+6\left(\frac{m\omega^2}{k}\right)^1-1=0$$

$$\alpha^3-5\alpha^2+6\alpha-1=0$$

Where $\left(\frac{m\omega^2}{k}\right)=\alpha$

Solving this third degree equation in α

$$\alpha=0.198, 1.554, 3.246$$

$$0.198=\frac{m\omega^2}{k}\Rrightarrow\Rrightarrow\Rrightarrow\ \omega_1=0.445\sqrt{\frac{k}{m}}$$

$$1.554=\frac{m\omega^2}{k}\Rrightarrow\Rrightarrow\Rrightarrow\ \omega_2=1.246\sqrt{\frac{k}{m}}$$

$$3.246=\frac{m\omega^2}{k}\Rrightarrow\Rrightarrow\Rrightarrow\ \omega_3=1.802\sqrt{\frac{k}{m}}$$

When $\omega=\omega_1=0.445\sqrt{\frac{k}{m}}$, Eqn (1) becomes

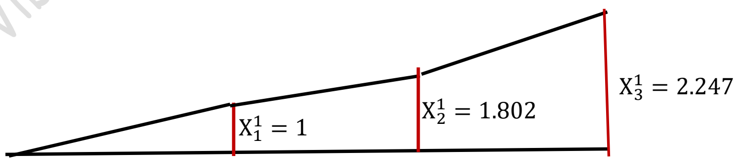
$$\begin{bmatrix}2k-\omega_1{}^2m & -k & 0\\ -k & 2k-\omega_1{}^2m & -k\\ 0 & -k & k-\omega_1{}^2m\end{bmatrix}\begin{Bmatrix}X_1{}^1\\ X_2{}^1\\ X_3{}^1\end{Bmatrix}=\begin{Bmatrix}0\\0\\0\end{Bmatrix}$$

$$\begin{bmatrix}1.802k & -k & 0\\ -k & 1.802k & -k\\ 0 & -k & 0.802k\end{bmatrix}\begin{Bmatrix}X_1{}^1\\ X_2{}^1\\ X_3{}^1\end{Bmatrix}=\begin{Bmatrix}0\\0\\0\end{Bmatrix}$$

$$1.802k\,X_1{}^1-k\,X_2{}^1=0\Rrightarrow\Rrightarrow\ \boxed{\frac{X_2{}^1}{X_1{}^1}=1.802}$$

$$-k\,X_1{}^1+1.802k\,X_2{}^1-kX_3{}^1=0\Rrightarrow\Rrightarrow\ -k\,X_1{}^1+1.802\times(1.802k\,X_1{}^1)-kX_3{}^1=0$$

$$2.247k\,X_1{}^1-k\,X_3{}^1=0\Rrightarrow\Rrightarrow\ \boxed{\frac{X_3{}^1}{X_1{}^1}=2.247}$$

I mode shape	Normal modes or modal vectors
	$\{X\}^1=\begin{Bmatrix}X_1{}^1\\ X_2{}^1\\ X_3{}^1\end{Bmatrix}=X_1{}^1\begin{Bmatrix}1\\ 1.802\\ 2.247\end{Bmatrix}$

When $\omega = \omega_2 = 1.246\sqrt{\frac{k}{m}}$, Eqn (1) becomes

$$\begin{bmatrix} 2k-\omega_2{}^2m & -k & 0 \\ -k & 2k-\omega_2{}^2m & -k \\ 0 & -k & k-\omega_2{}^2m \end{bmatrix}\begin{Bmatrix} X_1{}^2 \\ X_2{}^2 \\ X_3{}^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

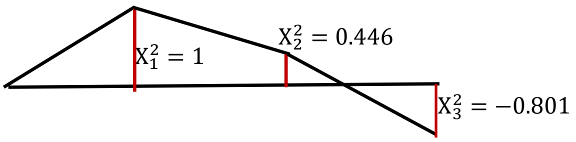
$$\begin{bmatrix} 0.446k & -k & 0 \\ -k & 0.446k & -k \\ 0 & -k & -0.554k \end{bmatrix}\begin{Bmatrix} X_1{}^2 \\ X_2{}^2 \\ X_3{}^2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

$$0.446k\,X_1{}^2 - k\,X_2{}^2 = 0 \Rrightarrow \left\|\frac{X_2{}^2}{X_1{}^2} = 0.446\right\|$$

$$-k\,X_1{}^2 + 0.446k\,X_2{}^2 - kX_3{}^2 = 0 \Rrightarrow -k\,X_1{}^2 + 0.446\times(0.446k\,X_1{}^2) - kX_3{}^2 = 0$$

$$-0.801k\,X_1{}^2 - k\,X_3{}^2 = 0 \Rrightarrow \left\|\frac{X_3{}^2}{X_1{}^2} = -0.801\right\|$$

II mode shape



Normal modes or modal vectors

$$\{X\}^2 = \begin{Bmatrix} X_1{}^2 \\ X_2{}^2 \\ X_3{}^2 \end{Bmatrix} = X_1{}^2\begin{Bmatrix} 1 \\ 0.446 \\ -0.801 \end{Bmatrix}$$

When $\omega = \omega_3 = 1.802\sqrt{\frac{k}{m}}$, Eqn (1) becomes

$$\begin{bmatrix} 2k-\omega_3{}^2m & -k & 0 \\ -k & 2k-\omega_3{}^2m & -k \\ 0 & -k & k-\omega_3{}^2m \end{bmatrix}\begin{Bmatrix} X_1{}^3 \\ X_2{}^3 \\ X_3{}^3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

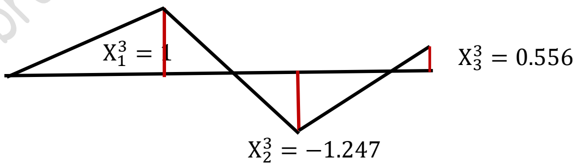
$$\begin{bmatrix} -1.247k & -k & 0 \\ -k & -1.247k & -k \\ 0 & -k & -2.247k \end{bmatrix}\begin{Bmatrix} X_1{}^3 \\ X_2{}^3 \\ X_3{}^3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

$$-1.247k\,X_1{}^3 - k\,X_2{}^3 = 0 \Rrightarrow \left\|\frac{X_2{}^3}{X_1{}^3} = -1.247\right\|$$

$$-k\,X_1{}^3 - 1.247k\,X_2{}^3 - kX_3{}^3 = 0 \Rrightarrow -k\,X_1{}^3 - 1.247\times(-1.247k\,X_1{}^3) - kX_3{}^3 = 0$$

$$0.556k\,X_1{}^3 - k\,X_3{}^3 = 0 \Rrightarrow \left\|\frac{X_3{}^3}{X_1{}^3} = 0.556\right\|$$

III mode shape



Normal modes or modal vectors

$$\{X\}^3 = \begin{Bmatrix} X_1{}^3 \\ X_2{}^3 \\ X_3{}^3 \end{Bmatrix} = X_1{}^3\begin{Bmatrix} 1 \\ -1.247 \\ 0.556 \end{Bmatrix}$$

Orthogonality of normal modes

The natural frequency ω_i and the corresponding modal vector $\{X\}^i$ satisfy the equation

$$([K] - \omega_i^2[M])\{X\}^i = \{0\}$$

$$\omega_i^2[M]\{X\}^i = [K]\{X\}^i ----(1)$$

If we consider another natural frequency ω_j and corresponding modal vectors $\{X\}^j$

$$\omega_j^2[M]\{X\}^j = [K]\{X\}^j --- (2)$$

Pre-multiplying Eq(1) by ${\{X\}^j}^T$ and Eq (2) by ${\{X\}^i}^T$

$$\omega_i^2{\{X\}^j}^T[M]\{X\}^i = {\{X\}^j}^T[K]\{X\}^i --- (3)$$

$$\omega_j^2{\{X\}^i}^T[M]\{X\}^j = {\{X\}^i}^T[K]\{X\}^j --- (4)$$

Since, $[K]$ and $[M]$ are symmetric matrices

$${\{X\}^j}^T[K]\{X\}^i = {\{X\}^i}^T[K]\{X\}^j \text{ and } {\{X\}^i}^T[M]\{X\}^j = {\{X\}^j}^T[M]\{X\}^i$$

$$\therefore Eq(3) \ggg\ggg \omega_i^2{\{X\}^j}^T[M]\{X\}^i = {\{X\}^i}^T[K]\{X\}^j --- (5)$$

$$\therefore Eq(4) \ggg\ggg \omega_j^2{\{X\}^j}^T[M]\{X\}^i = {\{X\}^i}^T[K]\{X\}^j --- (6)$$

Subtracting (6) from (5)

$$\omega_i^2{\{X\}^j}^T[M]\{X\}^i - \omega_j^2{\{X\}^j}^T[M]\{X\}^i = 0$$

$$(\omega_i^2 - \omega_j^2){\{X\}^j}^T[M]\{X\}^i = 0$$

But $\omega_i \neq \omega_j$

$${\{X\}^j}^T[M]\{X\}^i = 0;\ i \neq j$$

$$\therefore Eq(3) \ggg {\{X\}^j}^T[K]\{X\}^i = 0;\ \ i \neq j$$

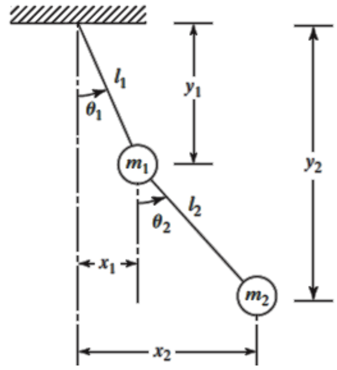
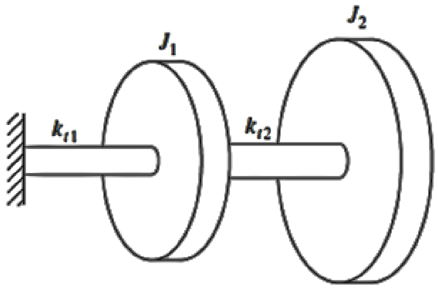
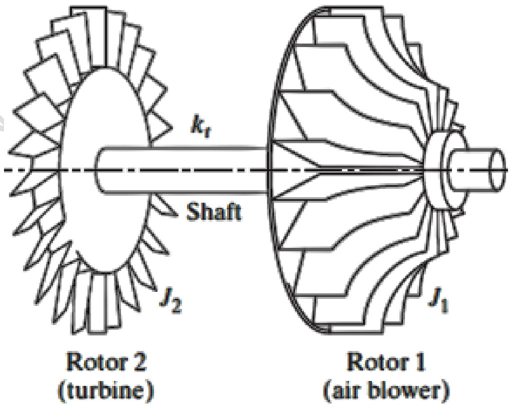
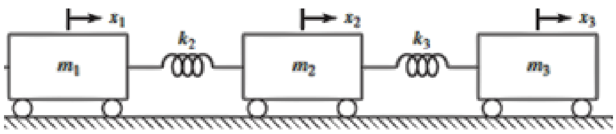
Thus, the modal vectors $\{X\}^i$ and $\{X\}^j$ are orthogonal with respect to the mass and stiffness matrices. If the modes are orthogonal, they will be linearly independent and can be used to describe the free vibration resulting from any initial conditions.

- Two vectors $\{X\}^i$ and $\{X\}^j$ are said to be orthogonal (perpendicular to each other if defined in 2 or 3 dimensional space), if ${\{X\}^i}^T\{X\}^j = 0$
- A vector $\{X\}^i$ is said to be normal, if its magnitude is unity .

$$\left|\{X\}^i\right|^2 = 1 \ \ or \ \ {\{X\}^i}^T\{X\}^i = 1.$$

- Two vectors $\{X\}^i$and $\{X\}^j$ are said to be ortho-normal, if they satisfy both orthogonality and normality conditions.

$${\{X\}^i}^T\{X\}^j = 0;\ \ {\{X\}^i}^T\{X\}^i = 1;\ \ \ \ {\{X\}^j}^T\{X\}^j = 1$$

Exercise 1 Derive the differential equations for the small amplitudes of oscillations for the double pendulum shown. Obtain the natural frequencies assuming $m_1 = m_2$ and $l_1 = l_2$	
Exercise 2 Derive the differential equations for the small amplitudes of oscillations for the two-rotor system shown. Obtain the natural frequencies and mode shapes assuming $J_2 = 2J_1$ and $k_{t2} = 2k_{t1}$	
Exercise 3 Derive the differential equations for the small amplitudes of oscillations for the two-rotor system shown. Obtain the natural frequencies and mode shapes assuming the mass moment of inertia of the rotors as $J_2 = 100\ kgm^2$, $J_1 = 50\ kgm^2$, diameter of the shaft $d = 10\ cm$, length f the shaft $l = 1\ m$, Modulus of rigidity of the shaft $G = 70\ GPa$	
Exercise 4 Determine the natural frequencies and mode shapes of the three degree of freedom system shown if $m_1 = 1\ kg$; and $m_2 = 2\ kg$; $m_3 = 3\ kg$ and $k_2 = 1 kN/m$; $k_3 = 2\ kN/m$	

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition